Самое начало

```
!pip install uv -q
!uv pip install --system "sglang[all] @ git+https://github.com/sgl-project/sg
!uv pip install --system openai httpx -q

import subprocess
import time

import sglang as sgl
print(f"SGLang установлен версии {sgl.__version__}")
```

```
◇[2K   ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m23.5/23.5 MB◇
◇[?25h◇[2mUsing Python 3.12.12 environment at: /usr◇[0m
◇[2K◇[2mResolved ◇[1m203 packages◇[0m ◇[2min 8.66s◇[0m◇[0m
◇[2K◇[2mPrepared ◇[1m76 packages◇[0m ◇[2min 24.57s◇[0m◇[0m
◇[2mUninstalled ◇[1m24 packages◇[0m ◇[2min 1.23s◇[0m◇[0m
◇[2K◇[2mInstalled ◇[1m76 packages◇[0m ◇[2min 302ms◇[0m◇[0m
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◇[32m+◇[39m ◇[1msglang-kernel◇[0m◇[2m==0.4.0◇[0m
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```
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 ◇[32m+◇[39m ◇[1mxgrammar◇[0m◇[2m==0.1.27◇[0m
SGLang установлен версии 0.5.6.post2
```

работа FSM и экстракшн сырого текста в json формат - вау

```
import json
import sglang as sgl
import time
# 1. Определяем промпт-программу с помощью декоратора
@sgl.function
def extract_character_info(s, description):
    # Системный промпт кэшируется в Radix Tree и не пересчитывается для каждо
    s += sgl.system("You are an expert data extractor. Extract the character 
    s += sgl.user(description)

    # 2. Магия FSM: Определяем строгий шаблон JSON
    # Модель физически не сможет сгенерировать текст, нарушающий этот regex
    regex_pattern = r"""\{\n  "name": "[\w\s]+",\n  "age": \d+,\n  "weapon":

    # Генерируем ответ, принудительно направляя вероятности токенов по regex-
    s += sgl.assistant(sgl.gen("json_output", regex=regex_pattern, temperatur

if __name__ == "__main__":
    print("Инициализация SGLang Runtime (загрузка весов в VRAM)...")
    # Запускаем локальный рантайм.
    # disable_cuda_graph=True иногда полезен в Colab для экономии памяти при 
    runtime = sgl.Runtime(model_path="Qwen/Qwen3.5-4B", dtype="bfloat16")
    sgl.set_default_backend(runtime)
```

```
    # Подготовим небольшой батч данных
    texts = [
        "Geralt of Rivia is a witcher. He is 95 years old and uses a silver sw
        "Luke Skywalker is 19 years old. He wields a blue lightsaber and figh
        "Arthur Pendragon, aged 35, pulled the legendary sword Excalibur from
    ]

    requests = [{"description": t} for t in texts]

    print("\n Запуск пакетной обработки (Batch FSM Extraction)...")
    start = time.perf_counter()

    # Отправляем батч. SGLang автоматически распараллелит запросы.
    states = extract_character_info.run_batch(requests)

    latency = time.perf_counter() - start

    print(f"\n Батч выполнен за: {latency:.2f} сек")
    print("-" * 40)

    for i, state in enumerate(states):
        print(f"Исходный текст: {texts[i]}")
        # Вытаскиваем сгенерированный JSON по ключу "json_output"
        raw_json = state["json_output"]

        # Проверяем, что это действительно парсится как JSON в Python
        parsed_dict = json.loads(raw_json)
        print(f"Извлеченный JSON: {json.dumps(parsed_dict, ensure_ascii=False

    # 3. ОСВОБОЖДАЕМ ПАМЯТЬ
    # Это критически важно для следующих шагов с сервером
    print("\n Остановка Runtime и очистка VRAM...")
    runtime.shutdown()
    import torch, gc
    gc.collect()
    torch.cuda.empty_cache()
    print("Шаг 2 завершен.")
```

```
WARNING:sglang.srt.configs.model_config:Transformers version 5.3.0 is used fo


Инициализация SGLang Runtime (загрузка весов в VRAM)...


WARNING:sglang.srt.server_args:Disabling overlap schedule since mamba no_buff



 Запуск пакетной обработки (Batch FSM Extraction)...
```

```
 Батч выполнен за: 2.63 сек
----------------------------------------
Исходный текст: Geralt of Rivia is a witcher. He is 95 years old and uses a si
Извлеченный JSON: {
  "name": "Geralt of Rivia",
  "age": 95,
  "weapon": "silver sword"
}

Исходный текст: Luke Skywalker is 19 years old. He wields a blue lightsaber an
Извлеченный JSON: {
  "name": "Luke Skywalker",
  "age": 19,
  "weapon": "blue lightsaber"
}

Исходный текст: Arthur Pendragon, aged 35, pulled the legendary sword Excalibu
Извлеченный JSON: {
  "name": "Arthur Pendragon",
  "age": 35,
  "weapon": "Excalibur"
}


 Остановка Runtime и очистка VRAM...
Шаг 2 завершен.
```

Продемонстрируем работу RadixAttention на большом контексте, и увидем, что если второй запрос относится к теме первого, то он просто подтягивает KV-cache из старого. Кэширование префиксов, чтобы префил второго реквеста был меньше

```
import sglang as sgl
import time

@sgl.function
def qa_bot(s, context, question):
    s += sgl.system(context)
    s += sgl.user(question)
    s += sgl.assistant(sgl.gen("answer", max_tokens=1024, temperature=0.9))

if __name__ == "__main__":
    print(" Поднимаем Runtime с защитой от OOM")

    # ИСПОЛЬЗУЕМ ЗАЩИТНЫЕ ПАРАМЕТРЫ ДЛЯ КОЛАБА
    runtime = sgl.Runtime(
        model_path="Qwen/Qwen3.5-4B",
        dtype="bfloat16",
```

```
        mem_fraction_static=0.8,   # Берем только 80% VRAM под кэш, оставляем
        chunked_prefill_size=4096   # Бьем длинный промпт на чанки по 4к токе
    )
    sgl.set_default_backend(runtime)

    # Имитируем огромный контекст 2546 на 4 = 10к
    long_context = """Efficient KV Cache Reuse with RadixAttention
SGLang programs can chain multiple generation calls and create parallel copies

Given the KV cache reuse opportunity, a key challenge in optimizing SGLang pro

This section introduces RadixAttention, a novel technique for automatic and sy

RadixAttention. A radix tree is a data structure that serves as a space-effic

In the continuous batching setting, we cannot evict nodes used by the current

Cache-aware scheduling. We define the cache hit rate as
number of cached prompt tokens
number of prompt tokens
. When there are many requests in the waiting queue, the order in which they a

Theorem 3.1.
For a batch of requests, we can achieve an optimal cache hit rate by visiting
≥
 the maximum request length. The longest-shared-prefix-first order is equivale

The proof is in Sec. A.3 (Appendix). In the online case, the DFS order will be

Distributed Cases. RadixAttention can be extended to multiple GPUs. For tenso

4 Efficient Constrained Decoding with Compressed Finite State Machine
In LM programs, users often want to constrain the model’s output to follow spe
as JSON schemas. This can improve controllability and robustness, and make the
parse. SGLang offers a regex argument to enforce such constraints using regula
are expressive enough for many practical scenarios. Existing systems support t
regular expression into a finite state machine (FSM) [54]. During decoding, th
FSM state, retrieve allowed tokens from the next states, and set the probabili
zero, decoding token by token. This token-by-token approach, however, is ineff
opportunities to decode multiple tokens at once. For example, the constant seq
" in Fig. 2 spans multiple tokens in the normal decoding process as shown in F
multiple decoding stages, even though there is only one valid next token when
the whole sequence can be decoded in a single step (i.e., forward pass). Howev
can only decode one token at a time because the lack of integration between th
runner in existing systems prevents multi-token processing, resulting in slow
SGLang overcomes this limitation by creating a fast constrained decoding runti
FSM. This runtime analyzes the FSM and compresses adjacent singular-transition
into single edges as demonstrated in Fig. 4 (b), allowing it to recognize when
```

decoded together. In Fig. 4 (d), multiple tokens on the compressed transition
one forward pass, which greatly accelerates the decoding process. It is also g
all regular expressions. More details on the background and implementation are

5 Efficient Endpoint Calling with API Speculative Execution

The previous sections introduced optimizations for open-weight models, which
to the model inference process. Additionally, SGLang works with API-access-onl
OpenAI's GPT-4. However, for these models, we can only call a black-box API en
This section introduces a new optimization for black-box API models that accel
and reduces the API cost of multi-call SGLang programs using speculative execu

pattern: s += context + "name:" + gen("name", stop="\n") + "job:" + gen("job"
stop="\n"). Naively, the two gen primitives correspond to two API calls, mean
needs to pay for the input token fee on the context twice. In SGLang, we can e
execution on the first call and let it continue the generation of a few more t
condition. The interpreter keeps the additional generation outputs and matches
later primitives. In certain cases, with careful prompt engineering, the model
template with high accuracy, saving us the latency and input costs of one API

6 Evaluation

We evaluate the performance of SGLang across diverse LLM workloads. Subsequent
ablation studies and case studies to demonstrate the effectiveness of specific
implemented in PyTorch [37] with custom CUDA kernels from FlashInfer [59] and

6.1 Setup

Models. We test dense Llama-2 models [49], sparse mixture of experts Mixtral m
models, the number of parameters ranges from 7 billion to 70 billion, and we u
Hardware. We run most experiments on AWS EC2 G5 instances, which are equipped
A10G GPUs (24GB). We run 7B models on a single A10G GPU and larger models on m
GPUs with tensor parallelism [44]. We run some additional experiments on A100G
Baselines. We compare SGLang against both high-level programming systems with
languages and default runtimes, as well as low-level inference engines with st
Completion APIs. Unless otherwise stated, we do not turn on optimizations that
computation results so that all systems compute the same results. The baseline
• Guidance[13], a language for controlling LLMs. We use Guidance v0.1.8 with T
• vLLM [23], a high-throughput inference engine. We use vLLM v0.2.5 and its de
.
• LMQL [4], a query language. We use LMQL v0.7.3 with Hugging Face Transforme
Workloads. We test the following: 5-shot MMLU [14] and 20-shot HellaSwag [61]
We decode one token for MMLU and use primitive select to select the answer wit
probability for HellaSwag. For the ReAct agent [57] and generative agents [36
traces from the original papers and replay them. We use the Tree-of-thought [5
problems and Skeleton-of-thought [33] for tip generation. We use LLM judges wi
chat with 4 turns, where the input of each turn is randomly sampled between 25
(256-512 tokens); DSPy retrieval-augmented generation (RAG) pipeline [20] in i
Metrics. We report two performance metrics: throughput and latency. For throug
sufficiently large batch of program instances to compute the maximum throughpu
number of program instances executed per second (programs per second, p/s). Fo
a single program at a time without batching and report the average latency fo
""" * 4

```
    print("\n ЗАПРОС 1 (Холодный старт - строим Radix Tree)...")
    start_1 = time.perf_counter()
    state_1 = qa_bot.run(context=long_context, question="О чем этот текст? Отв
    latency_1 = time.perf_counter() - start_1
    print(f" Время Запроса 1: {latency_1:.8f} сек")
    print(f"Ответ: {state_1['answer'].strip()}")

    print("\nЗАПРОС 2 (Radix Cache Hit)...")
    start_2 = time.perf_counter()
    state_2 = qa_bot.run(context=long_context, question="Какая технология упом
    latency_2 = time.perf_counter() - start_2
    print(f" Время Запроса 2: {latency_2:.8f} сек")
    print(f" Ответ: {state_2['answer'].strip()}")

    if latency_2 > 0:
        speedup = latency_1 / latency_2
        print(f"\n Итог: Второй запрос обработан в {speedup:.4f} раз быстрее!

    print("\n Очистка VRAM...")
    runtime.shutdown()
    import torch, gc
    gc.collect()
    torch.cuda.empty_cache()
    print(" Шаг 2.5 успешно завершен.")
```

```
WARNING:sglang.srt.configs.model_config:Transformers version 5.3.0 is used fo
WARNING:sglang.srt.server_args:Disabling overlap schedule since mamba no_buffe


 Поднимаем Runtime с защитой от OOM

 ЗАПРОС 1 (Холодный старт - строим Radix Tree)...
 Время Запроса 1: 0.00067395 сек
Ответ: Текст представляет собой техническую статью о системе SGLang и методах

<think>

</think>

Текст представляет собой техническую статью о системе SGLang и методах оптимиз

**Оптимизация**

ЗАПРОС 2 (Radix Cache Hit)...
 Время Запроса 2: 0.00038714 сек
 Ответ: Please provide the text content for analysis.

<think>
```

```
</think>

Основываясь на предоставленном тексте, упомянутой технологией является **Radix

Это новаторская техника, предложенная в работе над системой **SGLang**, котора

**Ключевые особенности этой технологии:**
*   **Структура данных:** Использует **radix tree** (дерево коевого типа) для
*   **Автоматизация:** В отличие от других систем, которые требуют ручного ко
*   **Управление памятью:** Внедряет политику эвикции (удаления) на основе LRU
*   **Градиентная оптимизация:** Включает алгоритм градиентной расписания, ко
*   **Совместимость:** Относится к техникам такого типа, как continuous batch

Также в тексте упоминаются другие связанные технологии и концепции:
*   **Compressed Finite State Machine (CFSM):** Для эффективного constrained
*   **Speculative Execution:** Используется для оптимизации вызовов API (черн
*   **Continuous batching:** Метод группировки запросов для ускорения inferen
*   **Tensor parallelism:** Распределение вычислений между несколькими GPU.

 Итог: Второй запрос обработан в 1.7408 раз быстрее!

 Очистка VRAM...
 Шаг 2.5 успешно завершен.
```

Посмотрим на метрики (любимое) SGlang инференса через докер и графану
как из гайда репы сглангa

upd: была попытка, но url страница лагала

```
🔷 1. Инициализация надежного Docker-демона...
✅ Docker работает стабильно.

🔷 2. Создание конфигурации Grafana и Prometheus...

⏳ 3. Запуск контейнеров мониторинга...
--- Контейнеры в работе ---
{.Names}   {.Status}   {.Ports}

⏳ 4. Поднимаем бэкенд SGLang (порт 30000)...
Ждем 35 секунд для загрузки весов модели в VRAM...

⏳ 5. Настройка внешнего туннеля (LocalTunnel)...

============================================================
🌐 ССЫЛКА НА GRAFANA:
👉 your url is: https://legal-lies-throw.loca.lt
============================================================
🔐 ПАРОЛЬ ДЛЯ ТУННЕЛЯ (Твой IP-адрес):
34.126.87.152
```

```
============================================================
🔑 Логин в Grafana: admin
🔑 Пароль в Grafana: admin

🚀 6. Запуск фонового генератора трафика...
✅ Готово! Переходи по ссылке выше.
```

Квантизируем модель опираясь на sglang/examples/usage /modelopt_quantize_and_export.py

на A100

```
# 2. Устанавливаем SGLang и библиотеку квантования NVIDIA ModelOpt
!pip install uv -q
!uv pip install --system "sglang[all] @ git+https://github.com/sgl-project/sgl
!uv pip install --system nvidia-modelopt[all] -q



print("◇ 1. Лечим конфликт библиотек (Protobuf)...")
# Устанавливаем стабильную версию protobuf, чтобы убрать ошибку MessageFactory
!pip install protobuf==3.20.3 -q

print(" DONE.")

◇[2K   ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m23.5/23.5 MB◇
◇[?25h◇[2mUsing Python 3.12.12 environment at: /usr◇[0m
◇[2K◇[2mResolved ◇[1m203 packages◇[0m ◇[2min 12.09s◇[0m◇[0m
◇[2K◇[2mPrepared ◇[1m76 packages◇[0m ◇[2min 22.28s◇[0m◇[0m
◇[2mUninstalled ◇[1m24 packages◇[0m ◇[2min 1.56s◇[0m◇[0m
◇[2K◇[2mInstalled ◇[1m76 packages◇[0m ◇[2min 2.55s◇[0m◇[0m
 ◇[32m+◇[39m ◇[1maddict◇[0m◇[2m==2.4.0◇[0m
 ◇[32m+◇[39m ◇[1mairportsdata◇[0m◇[2m==20260315◇[0m
 ◇[32m+◇[39m ◇[1manthropic◇[0m◇[2m==0.86.0◇[0m
 ◇[32m+◇[39m ◇[1mapache-tvm-ffi◇[0m◇[2m==0.1.9◇[0m
 ◇[32m+◇[39m ◇[1mav◇[0m◇[2m==16.1.0◇[0m
 ◇[31m-◇[39m ◇[1mblobfile◇[0m◇[2m==3.2.0◇[0m
 ◇[32m+◇[39m ◇[1mblobfile◇[0m◇[2m==3.0.0◇[0m
 ◇[32m+◇[39m ◇[1mbuild◇[0m◇[2m==1.4.0◇[0m
 ◇[32m+◇[39m ◇[1mcache-dit◇[0m◇[2m==1.3.0◇[0m
 ◇[32m+◇[39m ◇[1mcompressed-tensors◇[0m◇[2m==0.13.0◇[0m
 ◇[31m-◇[39m ◇[1mcuda-python◇[0m◇[2m==12.9.4◇[0m
 ◇[32m+◇[39m ◇[1mcuda-python◇[0m◇[2m==12.9.0◇[0m
 ◇[32m+◇[39m ◇[1mdiskcache◇[0m◇[2m==5.6.3◇[0m
 ◇[32m+◇[39m ◇[1mflash-attn-4◇[0m◇[2m==4.0.0b4◇[0m
 ◇[32m+◇[39m ◇[1mflashinfer-cubin◇[0m◇[2m==0.6.6◇[0m
 ◇[32m+◇[39m ◇[1mflashinfer-python◇[0m◇[2m==0.6.6◇[0m
 ◇[32m+◇[39m ◇[1mgguf◇[0m◇[2m==0.18.0◇[0m
 ◇[32m+◇[39m ◇[1mgrpcio-health-checking◇[0m◇[2m==1.78.0◇[0m
```

```
◇[32m+◇[39m ◇[1mgrpcio-reflection◇[0m◇[2m==1.78.0◇[0m
◇[31m-◇[39m ◇[1mimageio◇[0m◇[2m==2.37.3◇[0m
◇[32m+◇[39m ◇[1mimageio◇[0m◇[2m==2.36.0◇[0m
◇[31m-◇[39m ◇[1mimageio-ffmpeg◇[0m◇[2m==0.6.0◇[0m
◇[32m+◇[39m ◇[1mimageio-ffmpeg◇[0m◇[2m==0.5.1◇[0m
◇[32m+◇[39m ◇[1minteregular◇[0m◇[2m==0.3.3◇[0m
◇[32m+◇[39m ◇[1mjedi◇[0m◇[2m==0.19.2◇[0m
◇[32m+◇[39m ◇[1mllguidance◇[0m◇[2m==0.7.30◇[0m
◇[32m+◇[39m ◇[1mloguru◇[0m◇[2m==0.7.3◇[0m
◇[32m+◇[39m ◇[1mmistral-common◇[0m◇[2m==1.10.0◇[0m
◇[32m+◇[39m ◇[1mmodelscope◇[0m◇[2m==1.35.1◇[0m
◇[31m-◇[39m ◇[1mmoviepy◇[0m◇[2m==1.0.3◇[0m
◇[32m+◇[39m ◇[1mmoviepy◇[0m◇[2m==2.2.1◇[0m
◇[32m+◇[39m ◇[1mmsgspec◇[0m◇[2m==0.20.0◇[0m
◇[32m+◇[39m ◇[1mninja◇[0m◇[2m==1.13.0◇[0m
◇[32m+◇[39m ◇[1mnvidia-cudnn-frontend◇[0m◇[2m==1.20.0◇[0m
◇[32m+◇[39m ◇[1mnvidia-cutlass-dsl◇[0m◇[2m==4.4.2◇[0m
◇[32m+◇[39m ◇[1mnvidia-cutlass-dsl-libs-base◇[0m◇[2m==4.4.2◇[0m
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◇[31m-◇[39m ◇[1mopenai◇[0m◇[2m==2.28.0◇[0m
◇[32m+◇[39m ◇[1mopenai◇[0m◇[2m==2.6.1◇[0m
◇[32m+◇[39m ◇[1mopenai-harmony◇[0m◇[2m==0.0.4◇[0m
◇[31m-◇[39m ◇[1mopencv-python-headless◇[0m◇[2m==4.13.0.92◇[0m
◇[32m+◇[39m ◇[1mopencv-python-headless◇[0m◇[2m==4.10.0.84◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-api◇[0m◇[2m==1.38.0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-api◇[0m◇[2m==1.40.0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-exporter-otlp◇[0m◇[2m==1.40.0◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-exporter-otlp-proto-common◇[0m◇[2m==1.38.0◇
◇[32m+◇[39m ◇[1mopentelemetry-exporter-otlp-proto-common◇[0m◇[2m==1.40.0◇
◇[32m+◇[39m ◇[1mopentelemetry-exporter-otlp-proto-grpc◇[0m◇[2m==1.40.0◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-exporter-otlp-proto-http◇[0m◇[2m==1.38.0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-exporter-otlp-proto-http◇[0m◇[2m==1.40.0◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-proto◇[0m◇[2m==1.38.0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-proto◇[0m◇[2m==1.40.0◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-sdk◇[0m◇[2m==1.38.0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-sdk◇[0m◇[2m==1.40.0◇[0m
◇[31m-◇[39m ◇[1mopentelemetry-semantic-conventions◇[0m◇[2m==0.59b0◇[0m
◇[32m+◇[39m ◇[1mopentelemetry-semantic-conventions◇[0m◇[2m==0.61b0◇[0m
◇[32m+◇[39m ◇[1moutlines◇[0m◇[2m==0.1.11◇[0m
◇[32m+◇[39m ◇[1moutlines-core◇[0m◇[2m==0.1.26◇[0m
◇[32m+◇[39m ◇[1mpartial-json-parser◇[0m◇[2m==0.2.1.1.post7◇[0m
◇[31m-◇[39m ◇[1mprotobuf◇[0m◇[2m==5.29.6◇[0m
◇[32m+◇[39m ◇[1mprotobuf◇[0m◇[2m==6.33.6◇[0m
◇[32m+◇[39m ◇[1mpy-spy◇[0m◇[2m==0.4.1◇[0m
◇[32m+◇[39m ◇[1mpybase64◇[0m◇[2m==1.4.3◇[0m
◇[32m+◇[39m ◇[1mpycountry◇[0m◇[2m==26.2.16◇[0m
◇[32m+◇[39m ◇[1mpydantic-extra-types◇[0m◇[2m==2.11.1◇[0m
```

```
 ◇[32m+◇[39m ◇[1mpyproject-hooks◇[0m◇[2m==1.2.0◇[0m
 ◇[31m-◇[39m ◇[1mpyyaml◇[0m◇[2m==6.0.3◇[0m
 ◇[32m+◇[39m ◇[1mpyyaml◇[0m◇[2m==6.0.1◇[0m
 ◇[32m+◇[39m ◇[1mquack-kernels◇[0m◇[2m==0.3.5◇[0m
 ◇[32m+◇[39m ◇[1mremote-pdb◇[0m◇[2m==2.1.0◇[0m
 ◇[32m+◇[39m ◇[1mrunai-model-streamer◇[0m◇[2m==0.15.8◇[0m
 ◇[32m+◇[39m ◇[1msetproctitle◇[0m◇[2m==1.3.7◇[0m
 ◇[32m+◇[39m ◇[1msglang◇[0m◇[2m==0.5.6.post2 (from git+https://github.com/s
 ◇[32m+◇[39m ◇[1msglang-kernel◇[0m◇[2m==0.4.0◇[0m
 ◇[32m+◇[39m ◇[1msmg-grpc-proto◇[0m◇[2m==0.4.3◇[0m
 ◇[32m+◇[39m ◇[1msmg-grpc-servicer◇[0m◇[2m==0.5.0◇[0m
 ◇[32m+◇[39m ◇[1mst-attn◇[0m◇[2m==0.0.7◇[0m
 ◇[31m-◇[39m ◇[1mtimm◇[0m◇[2m==1.0.25◇[0m
 ◇[32m+◇[39m ◇[1mtimm◇[0m◇[2m==1.0.16◇[0m
 ◇[31m-◇[39m ◇[1mtorch◇[0m◇[2m==2.10.0+cu128◇[0m
 ◇[32m+◇[39m ◇[1mtorch◇[0m◇[2m==2.9.1◇[0m
 ◇[32m+◇[39m ◇[1mtorch-c-dlpack-ext◇[0m◇[2m==0.1.5◇[0m
 ◇[32m+◇[39m ◇[1mtorch-memory-saver◇[0m◇[2m==0.0.9◇[0m
 ◇[31m-◇[39m ◇[1mtorchao◇[0m◇[2m==0.10.0◇[0m
 ◇[32m+◇[39m ◇[1mtorchao◇[0m◇[2m==0.9.0◇[0m
 ◇[31m-◇[39m ◇[1mtorchaudio◇[0m◇[2m==2.10.0+cu128◇[0m
 ◇[32m+◇[39m ◇[1mtorchaudio◇[0m◇[2m==2.9.1◇[0m
 ◇[31m-◇[39m ◇[1mtorchcodec◇[0m◇[2m==0.10.0+cu128◇[0m
 ◇[32m+◇[39m ◇[1mtorchcodec◇[0m◇[2m==0.9.1◇[0m
 ◇[31m-◇[39m ◇[1mtorchvision◇[0m◇[2m==0.25.0+cu128◇[0m
 ◇[32m+◇[39m ◇[1mtorchvision◇[0m◇[2m==0.24.1◇[0m
 ◇[31m-◇[39m ◇[1mtransformers◇[0m◇[2m==5.0.0◇[0m
 ◇[32m+◇[39m ◇[1mtransformers◇[0m◇[2m==5.3.0◇[0m
 ◇[32m+◇[39m ◇[1mtrimesh◇[0m◇[2m==4.11.4◇[0m
 ◇[31m-◇[39m ◇[1mtriton◇[0m◇[2m==3.6.0◇[0m
 ◇[32m+◇[39m ◇[1mtriton◇[0m◇[2m==3.5.1◇[0m
 ◇[32m+◇[39m ◇[1mvsa◇[0m◇[2m==0.0.4◇[0m
 ◇[32m+◇[39m ◇[1mxatlas◇[0m◇[2m==0.0.11◇[0m
 ◇[32m+◇[39m ◇[1mxgrammar◇[0m◇[2m==0.1.27◇[0m
◇ 1. Лечим конфликт библиотек (Protobuf)...
◇[2K   ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m162.1/162.1 kB
◇[?25h◇[31mERROR: pip's dependency resolver does not currently take into acco
grpcio-reflection 1.78.0 requires protobuf<7.0.0,>=6.31.1, but you have proto
opentelemetry-proto 1.40.0 requires protobuf<7.0,>=5.0, but you have protobuf
smg-grpc-proto 0.4.3 requires protobuf>=5.26.0, but you have protobuf 3.20.3 w
onnx 1.19.1 requires protobuf>=4.25.1, but you have protobuf 3.20.3 which is i
grpcio-health-checking 1.78.0 requires protobuf<7.0.0,>=6.31.1, but you have p
ydf 0.15.0 requires protobuf<7.0.0,>=5.29.1, but you have protobuf 3.20.3 whic
wandb 0.25.1 requires protobuf!=5.28.0,!=5.29.0,<7,>4.21.0, but you have proto
google-api-core 2.30.0 requires protobuf<7.0.0,>=4.25.8, but you have protobu
grain 0.2.16 requires protobuf>=5.28.3, but you have protobuf 3.20.3 which is
google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you have ope
google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have ope
```

```
google-adk 1.27.1 requires PyYAML<7.0.0,>=6.0.2, but you have pyyaml 6.0.1 whi
grpcio-status 1.71.2 requires protobuf<6.0dev,>=5.26.1, but you have protobuf
tensorflow-metadata 1.17.3 requires protobuf>=4.25.2; python_version >= "3.11
◇[0m DONE.
```

```
# 1. Монтируем Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

Этот скрипт качает 7 ГБ оригинальных весов, сожмет до FP8 и сложит мне на Диск.

```
import os
import torch
from sglang.srt.configs.device_config import DeviceConfig
from sglang.srt.configs.load_config import LoadConfig
from sglang.srt.configs.model_config import ModelConfig
from sglang.srt.distributed.parallel_state import init_distributed_environmen
from sglang.srt.model_loader.loader import get_model_loader

print("⚙ Настройка окружения ModelOpt...")
if not torch.distributed.is_initialized():
    os.environ["RANK"] = "0"
    os.environ["WORLD_SIZE"] = "1"
    os.environ["MASTER_ADDR"] = "localhost"
    os.environ["MASTER_PORT"] = "12355"
    os.environ["LOCAL_RANK"] = "0"

    init_distributed_environment(world_size=1, rank=0, local_rank=0, backend=
    initialize_model_parallel(tensor_model_parallel_size=1, pipeline_model_pa

# ВОЗВРАЩАЕМСЯ НА СТАБИЛЬНУЮ КЛАССИЧЕСКУЮ АРХИТЕКТУРУ
model_path = "Qwen/Qwen2.5-7B-Instruct"
export_dir = "/content/drive/MyDrive/Qwen2.5-7B-FP8-ModelOpt"
os.makedirs(export_dir, exist_ok=True)

model_config = ModelConfig(
    model_path=model_path,
    quantization="modelopt_fp8",
    trust_remote_code=True,
)

load_config = LoadConfig(modelopt_export_path=export_dir)
device_config = DeviceConfig(device="cuda")

print(f"\n🚀 ЗАПУСК КВАНТОВАНИЯ НА A100: {model_path} -> FP8")
print(f"📁 Целевая папка (Google Drive): {export_dir}")
```

```
print("Процесс пошел. Можешь идти отдыхать!")

try:
    # Запуск движка компрессии
    model_loader = get_model_loader(load_config, model_config)
    model_loader.load_model(model_config=model_config, device_config=device_c

    print("\n✅ КВАНТОВАНИЕ ЗАВЕРШЕНО УСПЕШНО!")
    print("\n--- Содержимое папки на Google Drive ---")
    get_ipython().system(f'ls -lh "{export_dir}"')

except Exception as e:
    print(f"\n❌ Ошибка во время квантования: {e}")
```

```
⚙ Настройка окружения ModelOpt...


<frozen importlib._bootstrap_external>:1301: FutureWarning: The cuda.cudart mo
<frozen importlib._bootstrap_external>:1301: FutureWarning: The cuda.nvrtc mod



config.json:   0%|          | 0.00/663 [00:00<?, ?B/s]



generation_config.json:   0%|          | 0.00/243 [00:00<?, ?B/s]



🚀 ЗАПУСК КВАНТОВАНИЯ НА A100: Qwen/Qwen2.5-7B-Instruct -> FP8
📁 Целевая папка (Google Drive): /content/drive/MyDrive/Qwen2.5-7B-FP8-ModelO
Процесс пошел. Можешь идти отдыхать!


`torch_dtype` is deprecated! Use `dtype` instead!
`torch_dtype` is deprecated! Use `dtype` instead!



model.safetensors.index.json: 0.00B [00:00, ?B/s]



Fetching 4 files:   0%|          | 0/4 [00:00<?, ?it/s]
```

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model-00002-of-00004.safetensors:   0%|          | 0.00/3.86G [00:00<?, ?B/s]

model-00003-of-00004.safetensors:   0%|          | 0.00/3.86G [00:00<?, ?B/s]

model-00004-of-00004.safetensors:   0%|          | 0.00/3.56G [00:00<?, ?B/s]

model-00001-of-00004.safetensors:   0%|          | 0.00/3.95G [00:00<?, ?B/s]

Loading checkpoint shards:   0%|          | 0/4 [00:00<?, ?it/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

README.md: 0.00B [00:00, ?B/s]

Registered <class 'transformers.models.qwen2.modeling_qwen2.Qwen2Attention'> t
Inserted 703 quantizers

/usr/local/lib/python3.12/dist-packages/modelopt/torch/quantization/nn/modules
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/modelopt/torch/quantization/nn/modules
  warnings.warn(
100%|██████████| 15/15 [00:20<00:00,  1.38s/it]
```

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model.layers.0.self_attn.q_proj.input_quantizer
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model.layers.0.self_attn.q_proj.weight_quantizer
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model.layers.0.mlp.up_proj.output_quantizer
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model.layers.7.mlp.down_proj.input_quantizer
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model.layers.12.self_attn.q_proj.input_quantizer
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model.layers.13.mlp.up_proj.input_quantizer
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model.layers.13.mlp.up_proj.weight_quantizer
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model.layers.13.mlp.down_proj.output_quantizer
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model.layers.20.mlp.up_proj.output_quantizer
model.layers.20.mlp.up_proj.weight_quantizer
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model.layers.20.mlp.down_proj.output_quantizer
model.layers.20.mlp.down_proj.weight_quantizer
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model.layers.26.mlp.gate_proj.weight_quantizer
model.layers.26.mlp.up_proj.input_quantizer
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model.layers.27.mlp.down_proj.weight_quantizer
lm_head.input_quantizer
lm_head.output_quantizer
lm_head.weight_quantizer
703 TensorQuantizers found in model


Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
/usr/local/lib/python3.12/dist-packages/modelopt/torch/export/unified_export_
  weight_scaling_factor = torch.tensor(weight_quantizer.amax / weight_quantiz
```

✅ КВАНТОВАНИЕ ЗАВЕРШЕНО УСПЕШНО!

```
--- Содержимое папки на Google Drive ---
total 8.2G
-rw------- 1 root root  605 Mar 23 01:06 added_tokens.json
```

```
-rw------- 1 root root 2.5K Mar 23 01:06 chat_template.jinja
-rw------- 1 root root 2.2K Mar 23 01:06 config.json
-rw------- 1 root root  243 Mar 23 01:06 generation_config.json
-rw------- 1 root root  239 Mar 23 01:06 hf_quant_config.json
-rw------- 1 root root 1.6M Mar 23 01:06 merges.txt
-rw------- 1 root root 4.7G Mar 23 01:06 model-00001-of-00002.safetensors
-rw------- 1 root root 3.5G Mar 23 01:06 model-00002-of-00002.safetensors
-rw------- 1 root root  61K Mar 23 01:06 model.safetensors.index.json
-rw------- 1 root root  613 Mar 23 01:06 special_tokens_map.json
-rw------- 1 root root 4.6K Mar 23 01:06 tokenizer_config.json
-rw------- 1 root root  11M Mar 23 01:06 tokenizer.json
-rw------- 1 root root 2.7M Mar 23 01:06 vocab.json
```

```
✅ КВАНТОВАНИЕ ЗАВЕРШЕНО УСПЕШНО!

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total 8.2G
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-rw------- 1 root root  61K Mar 23 01:06 model.safetensors.index.json
-rw------- 1 root root  613 Mar 23 01:06 special_tokens_map.json
-rw------- 1 root root 4.6K Mar 23 01:06 tokenizer_config.json
-rw------- 1 root root  11M Mar 23 01:06 tokenizer.json
-rw------- 1 root root 2.7M Mar 23 01:06 vocab.json
```

image.png

Квантизируем квена 2.5 32B

для него требуется не самый новый protobuf хотелось кванитизовать 3.5 квены, но всё ломает Gated Delta Net

```
!pip install uv -q
!uv pip install --system "sglang[all] @ git+https://github.com/sgl-project/sg
!uv pip install --system nvidia-modelopt[all] -q
!pip install protobuf==3.20.3 -q
```

```
◇[2K   ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m23.5/23.5 MB◇
◇[?25h◇[2mUsing Python 3.12.12 environment at: /usr◇[0m
◇[2K◇[2mResolved ◇[1m203 packages◇[0m ◇[2min 12.35s◇[0m◇[0m
◇[2K◇[2mPrepared ◇[1m76 packages◇[0m ◇[2min 27.10s◇[0m◇[0m
◇[2mUninstalled ◇[1m24 packages◇[0m ◇[2min 1.13s◇[0m◇[0m
◇[2K◇[2mInstalled ◇[1m76 packages◇[0m ◇[2min 296ms◇[0m◇[0m
```

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```

```
 ◇[32m+◇[39m ◇[1mtriton◇[0m◇[2m==3.5.1◇[0m
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 ◇[32m+◇[39m ◇[1mxgrammar◇[0m◇[2m==0.1.27◇[0m
◇[2K   ◇[90m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━◇[0m ◇[32m162.1/162.1 kB
◇[?25h◇[31mERROR: pip's dependency resolver does not currently take into acco
grpcio-reflection 1.78.0 requires protobuf<7.0.0,>=6.31.1, but you have protob
opentelemetry-proto 1.40.0 requires protobuf<7.0,>=5.0, but you have protobuf
smg-grpc-proto 0.4.3 requires protobuf>=5.26.0, but you have protobuf 3.20.3 w
onnx 1.19.1 requires protobuf>=4.25.1, but you have protobuf 3.20.3 which is i
grpcio-health-checking 1.78.0 requires protobuf<7.0.0,>=6.31.1, but you have p
ydf 0.15.0 requires protobuf<7.0.0,>=5.29.1, but you have protobuf 3.20.3 whic
wandb 0.25.1 requires protobuf!=5.28.0,!=5.29.0,<7,>4.21.0, but you have proto
google-api-core 2.30.0 requires protobuf<7.0.0,>=4.25.8, but you have protobuf
grain 0.2.16 requires protobuf>=5.28.3, but you have protobuf 3.20.3 which is
google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you have ope
google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have ope
google-adk 1.27.1 requires PyYAML<7.0.0,>=6.0.2, but you have pyyaml 6.0.1 whi
grpcio-status 1.71.2 requires protobuf<6.0dev,>=5.26.1, but you have protobuf
tensorflow-metadata 1.17.3 requires protobuf>=4.25.2; python_version >= "3.11
◇[0m
```

```
# 1. Монтируем Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os
import torch
from sglang.srt.configs.device_config import DeviceConfig
from sglang.srt.configs.load_config import LoadConfig
from sglang.srt.configs.model_config import ModelConfig
from sglang.srt.distributed.parallel_state import init_distributed_environmen
from sglang.srt.model_loader.loader import get_model_loader

print("⚙ Настройка окружения ModelOpt...")
if not torch.distributed.is_initialized():
    os.environ["RANK"] = "0"
    os.environ["WORLD_SIZE"] = "1"
    os.environ["MASTER_ADDR"] = "localhost"
    os.environ["MASTER_PORT"] = "12355"
    os.environ["LOCAL_RANK"] = "0"

    init_distributed_environment(world_size=1, rank=0, local_rank=0, backend=
    initialize_model_parallel(tensor_model_parallel_size=1, pipeline_model_pa

# БЕРЕМ МОДЕЛЬ НА 32 МИЛЛИАРДА ПАРАМЕТРОВ
model_path = "Qwen/Qwen2.5-32B-Instruct"
```

```
# СОХРАНЯЕМ НА ЛОКАЛЬНЫЙ ДИСК КОЛАБА (на Google Диск не влезет!)
export_dir = "/content/drive/Qwen2.5-32B-FP8-ModelOpt"
os.makedirs(export_dir, exist_ok=True)

model_config = ModelConfig(
    model_path=model_path,
    quantization="modelopt_fp8",
    trust_remote_code=True,
)

load_config = LoadConfig(modelopt_export_path=export_dir)
device_config = DeviceConfig(device="cuda")

print(f"\n🚀 ЗАПУСК КВАНТОВАНИЯ НА A100 (80GB): {model_path} -> FP8")
print(f"📁 Целевая папка (Локальный диск Colab): {export_dir}")
print("Это займет время (скачивание 64 ГБ весов + компрессия). Ждем...")

try:
    # Запуск движка компрессии
    model_loader = get_model_loader(load_config, model_config)
    model_loader.load_model(model_config=model_config, device_config=device_c

    print("\n✅ КВАНТОВАНИЕ 32B-МОДЕЛИ ЗАВЕРШЕНО УСПЕШНО!")
    print("\n--- Содержимое папки с готовой моделью ---")
    get_ipython().system(f'ls -lh "{export_dir}"')

except Exception as e:
    print(f"\n❌ Ошибка во время квантования: {e}")
```

```
⚙ Настройка окружения ModelOpt...


<frozen importlib._bootstrap_external>:1301: FutureWarning: The cuda.cudart mo
<frozen importlib._bootstrap_external>:1301: FutureWarning: The cuda.nvrtc mod



config.json:   0%|          | 0.00/663 [00:00<?, ?B/s]



generation_config.json:   0%|          | 0.00/242 [00:00<?, ?B/s]



🚀 ЗАПУСК КВАНТОВАНИЯ НА A100 (80GB): Qwen/Qwen2.5-32B-Instruct -> FP8
📁 Целевая папка (Локальный диск Colab): /content/Qwen2.5-32B-FP8-ModelOpt
```

```
Это займет время (скачивание 64 ГБ весов + компрессия). Ждем...

`torch_dtype` is deprecated! Use `dtype` instead!
`torch_dtype` is deprecated! Use `dtype` instead!


model.safetensors.index.json: 0.00B [00:00, ?B/s]


Fetching 17 files:   0%|          | 0/17 [00:00<?, ?it/s]


model-00006-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00007-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00005-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00001-of-00017.safetensors:   0%|          | 0.00/3.92G [00:00<?, ?B/s]


model-00003-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00002-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00004-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]


model-00008-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]
```

```
model-00009-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00010-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00011-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00012-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00013-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00014-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00015-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00016-of-00017.safetensors:   0%|          | 0.00/3.90G [00:00<?, ?B/s]

model-00017-of-00017.safetensors:   0%|          | 0.00/3.10G [00:00<?, ?B/s]

Loading checkpoint shards:   0%|          | 0/17 [00:00<?, ?it/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]
```

```
merges.txt: 0.00B [00:00, ?B/s]


tokenizer.json: 0.00B [00:00, ?B/s]


README.md: 0.00B [00:00, ?B/s]

Registered <class 'transformers.models.qwen2.modeling_qwen2.Qwen2Attention'> t

/usr/local/lib/python3.12/dist-packages/modelopt/torch/quantization/nn/modules
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/modelopt/torch/quantization/nn/modules
  warnings.warn(

Inserted 1603 quantizers

100%|██████████| 15/15 [01:22<00:00,  5.47s/it]

model.layers.0.self_attn.q_proj.input_quantizer
model.layers.0.self_attn.q_proj.output_quantizer
model.layers.0.self_attn.q_proj.weight_quantizer
model.layers.0.self_attn.k_proj.input_quantizer
model.layers.0.self_attn.k_proj.output_quantizer
model.layers.0.self_attn.k_proj.weight_quantizer
model.layers.0.self_attn.v_proj.input_quantizer
model.layers.0.self_attn.v_proj.output_quantizer
model.layers.0.self_attn.v_proj.weight_quantizer
model.layers.0.self_attn.o_proj.input_quantizer
model.layers.0.self_attn.o_proj.output_quantizer
model.layers.0.self_attn.o_proj.weight_quantizer
model.layers.0.self_attn.q_bmm_quantizer
model.layers.0.self_attn.k_bmm_quantizer
model.layers.0.self_attn.v_bmm_quantizer
model.layers.0.self_attn.softmax_quantizer
model.layers.0.mlp.gate_proj.input_quantizer
model.layers.0.mlp.gate_proj.output_quantizer
model.layers.0.mlp.gate_proj.weight_quantizer
model.layers.0.mlp.up_proj.input_quantizer
model.layers.0.mlp.up_proj.output_quantizer
model.layers.0.mlp.up_proj.weight_quantizer
```

model.layers.0.mlp.down_proj.input_quantizer
model.layers.0.mlp.down_proj.output_quantizer
model.layers.0.mlp.down_proj.weight_quantizer
model.layers.1.self_attn.q_proj.input_quantizer
model.layers.1.self_attn.q_proj.output_quantizer
model.layers.1.self_attn.q_proj.weight_quantizer
model.layers.1.self_attn.k_proj.input_quantizer
model.layers.1.self_attn.k_proj.output_quantizer
model.layers.1.self_attn.k_proj.weight_quantizer
model.layers.1.self_attn.v_proj.input_quantizer
model.layers.1.self_attn.v_proj.output_quantizer
model.layers.1.self_attn.v_proj.weight_quantizer
model.layers.1.self_attn.o_proj.input_quantizer
model.layers.1.self_attn.o_proj.output_quantizer
model.layers.1.self_attn.o_proj.weight_quantizer
model.layers.1.self_attn.q_bmm_quantizer
model.layers.1.self_attn.k_bmm_quantizer
model.layers.1.self_attn.v_bmm_quantizer
model.layers.1.self_attn.softmax_quantizer
model.layers.1.mlp.gate_proj.input_quantizer
model.layers.1.mlp.gate_proj.output_quantizer
model.layers.1.mlp.gate_proj.weight_quantizer
model.layers.1.mlp.up_proj.input_quantizer
model.layers.1.mlp.up_proj.output_quantizer
model.layers.1.mlp.up_proj.weight_quantizer
model.layers.1.mlp.down_proj.input_quantizer
model.layers.1.mlp.down_proj.output_quantizer
model.layers.1.mlp.down_proj.weight_quantizer
model.layers.2.self_attn.q_proj.input_quantizer
model.layers.2.self_attn.q_proj.output_quantizer
model.layers.2.self_attn.q_proj.weight_quantizer
model.layers.2.self_attn.k_proj.input_quantizer
model.layers.2.self_attn.k_proj.output_quantizer
model.layers.2.self_attn.k_proj.weight_quantizer
model.layers.2.self_attn.v_proj.input_quantizer
model.layers.2.self_attn.v_proj.output_quantizer
model.layers.2.self_attn.v_proj.weight_quantizer
model.layers.2.self_attn.o_proj.input_quantizer
model.layers.2.self_attn.o_proj.output_quantizer
model.layers.2.self_attn.o_proj.weight_quantizer
model.layers.2.self_attn.q_bmm_quantizer
model.layers.2.self_attn.k_bmm_quantizer
model.layers.2.self_attn.v_bmm_quantizer
model.layers.2.self_attn.softmax_quantizer
model.layers.2.mlp.gate_proj.input_quantizer
model.layers.2.mlp.gate_proj.output_quantizer
model.layers.2.mlp.gate_proj.weight_quantizer
model.layers.2.mlp.up_proj.input_quantizer

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model.layers.2.mlp.up_proj.output_quantizer
model.layers.2.mlp.up_proj.weight_quantizer
model.layers.2.mlp.down_proj.input_quantizer
model.layers.2.mlp.down_proj.output_quantizer
model.layers.2.mlp.down_proj.weight_quantizer
model.layers.3.self_attn.q_proj.input_quantizer
model.layers.3.self_attn.q_proj.output_quantizer
model.layers.3.self_attn.q_proj.weight_quantizer
model.layers.3.self_attn.k_proj.input_quantizer
model.layers.3.self_attn.k_proj.output_quantizer
model.layers.3.self_attn.k_proj.weight_quantizer
model.layers.3.self_attn.v_proj.input_quantizer
model.layers.3.self_attn.v_proj.output_quantizer
model.layers.3.self_attn.v_proj.weight_quantizer
model.layers.3.self_attn.o_proj.input_quantizer
model.layers.3.self_attn.o_proj.output_quantizer
model.layers.3.self_attn.o_proj.weight_quantizer
model.layers.3.self_attn.q_bmm_quantizer
model.layers.3.self_attn.k_bmm_quantizer
model.layers.3.self_attn.v_bmm_quantizer
model.layers.3.self_attn.softmax_quantizer
model.layers.3.mlp.gate_proj.input_quantizer
model.layers.3.mlp.gate_proj.output_quantizer
model.layers.3.mlp.gate_proj.weight_quantizer
model.layers.3.mlp.up_proj.input_quantizer
model.layers.3.mlp.up_proj.output_quantizer
model.layers.3.mlp.up_proj.weight_quantizer
model.layers.3.mlp.down_proj.input_quantizer
model.layers.3.mlp.down_proj.output_quantizer
model.layers.3.mlp.down_proj.weight_quantizer
model.layers.4.self_attn.q_proj.input_quantizer
model.layers.4.self_attn.q_proj.output_quantizer
model.layers.4.self_attn.q_proj.weight_quantizer
model.layers.4.self_attn.k_proj.input_quantizer
model.layers.4.self_attn.k_proj.output_quantizer
model.layers.4.self_attn.k_proj.weight_quantizer
model.layers.4.self_attn.v_proj.input_quantizer
model.layers.4.self_attn.v_proj.output_quantizer
model.layers.4.self_attn.v_proj.weight_quantizer
model.layers.4.self_attn.o_proj.input_quantizer
model.layers.4.self_attn.o_proj.output_quantizer
model.layers.4.self_attn.o_proj.weight_quantizer
model.layers.4.self_attn.q_bmm_quantizer
model.layers.4.self_attn.k_bmm_quantizer
model.layers.4.self_attn.v_bmm_quantizer
model.layers.4.self_attn.softmax_quantizer
model.layers.4.mlp.gate_proj.input_quantizer
model.layers.4.mlp.gate_proj.output_quantizer
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model.layers.4.mlp.gate_proj.weight_quantizer
model.layers.4.mlp.up_proj.input_quantizer
model.layers.4.mlp.up_proj.output_quantizer
model.layers.4.mlp.up_proj.weight_quantizer
model.layers.4.mlp.down_proj.input_quantizer
model.layers.4.mlp.down_proj.output_quantizer
model.layers.4.mlp.down_proj.weight_quantizer
model.layers.5.self_attn.q_proj.input_quantizer
model.layers.5.self_attn.q_proj.output_quantizer
model.layers.5.self_attn.q_proj.weight_quantizer
model.layers.5.self_attn.k_proj.input_quantizer
model.layers.5.self_attn.k_proj.output_quantizer
model.layers.5.self_attn.k_proj.weight_quantizer
model.layers.5.self_attn.v_proj.input_quantizer
model.layers.5.self_attn.v_proj.output_quantizer
model.layers.5.self_attn.v_proj.weight_quantizer
model.layers.5.self_attn.o_proj.input_quantizer
model.layers.5.self_attn.o_proj.output_quantizer
model.layers.5.self_attn.o_proj.weight_quantizer
model.layers.5.self_attn.q_bmm_quantizer
model.layers.5.self_attn.k_bmm_quantizer
model.layers.5.self_attn.v_bmm_quantizer
model.layers.5.self_attn.softmax_quantizer
model.layers.5.mlp.gate_proj.input_quantizer
model.layers.5.mlp.gate_proj.output_quantizer
model.layers.5.mlp.gate_proj.weight_quantizer
model.layers.5.mlp.up_proj.input_quantizer
model.layers.5.mlp.up_proj.output_quantizer
model.layers.5.mlp.up_proj.weight_quantizer
model.layers.5.mlp.down_proj.input_quantizer
model.layers.5.mlp.down_proj.output_quantizer
model.layers.5.mlp.down_proj.weight_quantizer
model.layers.6.self_attn.q_proj.input_quantizer
model.layers.6.self_attn.q_proj.output_quantizer
model.layers.6.self_attn.q_proj.weight_quantizer
model.layers.6.self_attn.k_proj.input_quantizer
model.layers.6.self_attn.k_proj.output_quantizer
model.layers.6.self_attn.k_proj.weight_quantizer
model.layers.6.self_attn.v_proj.input_quantizer
model.layers.6.self_attn.v_proj.output_quantizer
model.layers.6.self_attn.v_proj.weight_quantizer
model.layers.6.self_attn.o_proj.input_quantizer
model.layers.6.self_attn.o_proj.output_quantizer
model.layers.6.self_attn.o_proj.weight_quantizer
model.layers.6.self_attn.q_bmm_quantizer
model.layers.6.self_attn.k_bmm_quantizer
model.layers.6.self_attn.v_bmm_quantizer
model.layers.6.self_attn.softmax_quantizer

model.layers.6.mlp.gate_proj.input_quantizer
model.layers.6.mlp.gate_proj.output_quantizer
model.layers.6.mlp.gate_proj.weight_quantizer
model.layers.6.mlp.up_proj.input_quantizer
model.layers.6.mlp.up_proj.output_quantizer
model.layers.6.mlp.up_proj.weight_quantizer
model.layers.6.mlp.down_proj.input_quantizer
model.layers.6.mlp.down_proj.output_quantizer
model.layers.6.mlp.down_proj.weight_quantizer
model.layers.7.self_attn.q_proj.input_quantizer
model.layers.7.self_attn.q_proj.output_quantizer
model.layers.7.self_attn.q_proj.weight_quantizer
model.layers.7.self_attn.k_proj.input_quantizer
model.layers.7.self_attn.k_proj.output_quantizer
model.layers.7.self_attn.k_proj.weight_quantizer
model.layers.7.self_attn.v_proj.input_quantizer
model.layers.7.self_attn.v_proj.output_quantizer
model.layers.7.self_attn.v_proj.weight_quantizer
model.layers.7.self_attn.o_proj.input_quantizer
model.layers.7.self_attn.o_proj.output_quantizer
model.layers.7.self_attn.o_proj.weight_quantizer
model.layers.7.self_attn.q_bmm_quantizer
model.layers.7.self_attn.k_bmm_quantizer
model.layers.7.self_attn.v_bmm_quantizer
model.layers.7.self_attn.softmax_quantizer
model.layers.7.mlp.gate_proj.input_quantizer
model.layers.7.mlp.gate_proj.output_quantizer
model.layers.7.mlp.gate_proj.weight_quantizer
model.layers.7.mlp.up_proj.input_quantizer
model.layers.7.mlp.up_proj.output_quantizer
model.layers.7.mlp.up_proj.weight_quantizer
model.layers.7.mlp.down_proj.input_quantizer
model.layers.7.mlp.down_proj.output_quantizer
model.layers.7.mlp.down_proj.weight_quantizer
model.layers.8.self_attn.q_proj.input_quantizer
model.layers.8.self_attn.q_proj.output_quantizer
model.layers.8.self_attn.q_proj.weight_quantizer
model.layers.8.self_attn.k_proj.input_quantizer
model.layers.8.self_attn.k_proj.output_quantizer
model.layers.8.self_attn.k_proj.weight_quantizer
model.layers.8.self_attn.v_proj.input_quantizer
model.layers.8.self_attn.v_proj.output_quantizer
model.layers.8.self_attn.v_proj.weight_quantizer
model.layers.8.self_attn.o_proj.input_quantizer
model.layers.8.self_attn.o_proj.output_quantizer
model.layers.8.self_attn.o_proj.weight_quantizer
model.layers.8.self_attn.q_bmm_quantizer
model.layers.8.self_attn.k_bmm_quantizer

model.layers.8.self_attn.v_bmm_quantizer
model.layers.8.self_attn.softmax_quantizer
model.layers.8.mlp.gate_proj.input_quantizer
model.layers.8.mlp.gate_proj.output_quantizer
model.layers.8.mlp.gate_proj.weight_quantizer
model.layers.8.mlp.up_proj.input_quantizer
model.layers.8.mlp.up_proj.output_quantizer
model.layers.8.mlp.up_proj.weight_quantizer
model.layers.8.mlp.down_proj.input_quantizer
model.layers.8.mlp.down_proj.output_quantizer
model.layers.8.mlp.down_proj.weight_quantizer
model.layers.9.self_attn.q_proj.input_quantizer
model.layers.9.self_attn.q_proj.output_quantizer
model.layers.9.self_attn.q_proj.weight_quantizer
model.layers.9.self_attn.k_proj.input_quantizer
model.layers.9.self_attn.k_proj.output_quantizer
model.layers.9.self_attn.k_proj.weight_quantizer
model.layers.9.self_attn.v_proj.input_quantizer
model.layers.9.self_attn.v_proj.output_quantizer
model.layers.9.self_attn.v_proj.weight_quantizer
model.layers.9.self_attn.o_proj.input_quantizer
model.layers.9.self_attn.o_proj.output_quantizer
model.layers.9.self_attn.o_proj.weight_quantizer
model.layers.9.self_attn.q_bmm_quantizer
model.layers.9.self_attn.k_bmm_quantizer
model.layers.9.self_attn.v_bmm_quantizer
model.layers.9.self_attn.softmax_quantizer
model.layers.9.mlp.gate_proj.input_quantizer
model.layers.9.mlp.gate_proj.output_quantizer
model.layers.9.mlp.gate_proj.weight_quantizer
model.layers.9.mlp.up_proj.input_quantizer
model.layers.9.mlp.up_proj.output_quantizer
model.layers.9.mlp.up_proj.weight_quantizer
model.layers.9.mlp.down_proj.input_quantizer
model.layers.9.mlp.down_proj.output_quantizer
model.layers.9.mlp.down_proj.weight_quantizer
model.layers.10.self_attn.q_proj.input_quantizer
model.layers.10.self_attn.q_proj.output_quantizer
model.layers.10.self_attn.q_proj.weight_quantizer
model.layers.10.self_attn.k_proj.input_quantizer
model.layers.10.self_attn.k_proj.output_quantizer
model.layers.10.self_attn.k_proj.weight_quantizer
model.layers.10.self_attn.v_proj.input_quantizer
model.layers.10.self_attn.v_proj.output_quantizer
model.layers.10.self_attn.v_proj.weight_quantizer
model.layers.10.self_attn.o_proj.input_quantizer
model.layers.10.self_attn.o_proj.output_quantizer
model.layers.10.self_attn.o_proj.weight_quantizer

model.layers.10.self_attn.q_bmm_quantizer
model.layers.10.self_attn.k_bmm_quantizer
model.layers.10.self_attn.v_bmm_quantizer
model.layers.10.self_attn.softmax_quantizer
model.layers.10.mlp.gate_proj.input_quantizer
model.layers.10.mlp.gate_proj.output_quantizer
model.layers.10.mlp.gate_proj.weight_quantizer
model.layers.10.mlp.up_proj.input_quantizer
model.layers.10.mlp.up_proj.output_quantizer
model.layers.10.mlp.up_proj.weight_quantizer
model.layers.10.mlp.down_proj.input_quantizer
model.layers.10.mlp.down_proj.output_quantizer
model.layers.10.mlp.down_proj.weight_quantizer
model.layers.11.self_attn.q_proj.input_quantizer
model.layers.11.self_attn.q_proj.output_quantizer
model.layers.11.self_attn.q_proj.weight_quantizer
model.layers.11.self_attn.k_proj.input_quantizer
model.layers.11.self_attn.k_proj.output_quantizer
model.layers.11.self_attn.k_proj.weight_quantizer
model.layers.11.self_attn.v_proj.input_quantizer
model.layers.11.self_attn.v_proj.output_quantizer
model.layers.11.self_attn.v_proj.weight_quantizer
model.layers.11.self_attn.o_proj.input_quantizer
model.layers.11.self_attn.o_proj.output_quantizer
model.layers.11.self_attn.o_proj.weight_quantizer
model.layers.11.self_attn.q_bmm_quantizer
model.layers.11.self_attn.k_bmm_quantizer
model.layers.11.self_attn.v_bmm_quantizer
model.layers.11.self_attn.softmax_quantizer
model.layers.11.mlp.gate_proj.input_quantizer
model.layers.11.mlp.gate_proj.output_quantizer
model.layers.11.mlp.gate_proj.weight_quantizer
model.layers.11.mlp.up_proj.input_quantizer
model.layers.11.mlp.up_proj.output_quantizer
model.layers.11.mlp.up_proj.weight_quantizer
model.layers.11.mlp.down_proj.input_quantizer
model.layers.11.mlp.down_proj.output_quantizer
model.layers.11.mlp.down_proj.weight_quantizer
model.layers.12.self_attn.q_proj.input_quantizer
model.layers.12.self_attn.q_proj.output_quantizer
model.layers.12.self_attn.q_proj.weight_quantizer
model.layers.12.self_attn.k_proj.input_quantizer
model.layers.12.self_attn.k_proj.output_quantizer
model.layers.12.self_attn.k_proj.weight_quantizer
model.layers.12.self_attn.v_proj.input_quantizer
model.layers.12.self_attn.v_proj.output_quantizer
model.layers.12.self_attn.v_proj.weight_quantizer
model.layers.12.self_attn.o_proj.input_quantizer

model.layers.12.self_attn.o_proj.output_quantizer
model.layers.12.self_attn.o_proj.weight_quantizer
model.layers.12.self_attn.q_bmm_quantizer
model.layers.12.self_attn.k_bmm_quantizer
model.layers.12.self_attn.v_bmm_quantizer
model.layers.12.self_attn.softmax_quantizer
model.layers.12.mlp.gate_proj.input_quantizer
model.layers.12.mlp.gate_proj.output_quantizer
model.layers.12.mlp.gate_proj.weight_quantizer
model.layers.12.mlp.up_proj.input_quantizer
model.layers.12.mlp.up_proj.output_quantizer
model.layers.12.mlp.up_proj.weight_quantizer
model.layers.12.mlp.down_proj.input_quantizer
model.layers.12.mlp.down_proj.output_quantizer
model.layers.12.mlp.down_proj.weight_quantizer
model.layers.13.self_attn.q_proj.input_quantizer
model.layers.13.self_attn.q_proj.output_quantizer
model.layers.13.self_attn.q_proj.weight_quantizer
model.layers.13.self_attn.k_proj.input_quantizer
model.layers.13.self_attn.k_proj.output_quantizer
model.layers.13.self_attn.k_proj.weight_quantizer
model.layers.13.self_attn.v_proj.input_quantizer
model.layers.13.self_attn.v_proj.output_quantizer
model.layers.13.self_attn.v_proj.weight_quantizer
model.layers.13.self_attn.o_proj.input_quantizer
model.layers.13.self_attn.o_proj.output_quantizer
model.layers.13.self_attn.o_proj.weight_quantizer
model.layers.13.self_attn.q_bmm_quantizer
model.layers.13.self_attn.k_bmm_quantizer
model.layers.13.self_attn.v_bmm_quantizer
model.layers.13.self_attn.softmax_quantizer
model.layers.13.mlp.gate_proj.input_quantizer
model.layers.13.mlp.gate_proj.output_quantizer
model.layers.13.mlp.gate_proj.weight_quantizer
model.layers.13.mlp.up_proj.input_quantizer
model.layers.13.mlp.up_proj.output_quantizer
model.layers.13.mlp.up_proj.weight_quantizer
model.layers.13.mlp.down_proj.input_quantizer
model.layers.13.mlp.down_proj.output_quantizer
model.layers.13.mlp.down_proj.weight_quantizer
model.layers.14.self_attn.q_proj.input_quantizer
model.layers.14.self_attn.q_proj.output_quantizer
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model.layers.14.self_attn.k_proj.input_quantizer
model.layers.14.self_attn.k_proj.output_quantizer
model.layers.14.self_attn.k_proj.weight_quantizer
model.layers.14.self_attn.v_proj.input_quantizer
model.layers.14.self_attn.v_proj.output_quantizer

model.layers.14.self_attn.v_proj.weight_quantizer
model.layers.14.self_attn.o_proj.input_quantizer
model.layers.14.self_attn.o_proj.output_quantizer
model.layers.14.self_attn.o_proj.weight_quantizer
model.layers.14.self_attn.q_bmm_quantizer
model.layers.14.self_attn.k_bmm_quantizer
model.layers.14.self_attn.v_bmm_quantizer
model.layers.14.self_attn.softmax_quantizer
model.layers.14.mlp.gate_proj.input_quantizer
model.layers.14.mlp.gate_proj.output_quantizer
model.layers.14.mlp.gate_proj.weight_quantizer
model.layers.14.mlp.up_proj.input_quantizer
model.layers.14.mlp.up_proj.output_quantizer
model.layers.14.mlp.up_proj.weight_quantizer
model.layers.14.mlp.down_proj.input_quantizer
model.layers.14.mlp.down_proj.output_quantizer
model.layers.14.mlp.down_proj.weight_quantizer
model.layers.15.self_attn.q_proj.input_quantizer
model.layers.15.self_attn.q_proj.output_quantizer
model.layers.15.self_attn.q_proj.weight_quantizer
model.layers.15.self_attn.k_proj.input_quantizer
model.layers.15.self_attn.k_proj.output_quantizer
model.layers.15.self_attn.k_proj.weight_quantizer
model.layers.15.self_attn.v_proj.input_quantizer
model.layers.15.self_attn.v_proj.output_quantizer
model.layers.15.self_attn.v_proj.weight_quantizer
model.layers.15.self_attn.o_proj.input_quantizer
model.layers.15.self_attn.o_proj.output_quantizer
model.layers.15.self_attn.o_proj.weight_quantizer
model.layers.15.self_attn.q_bmm_quantizer
model.layers.15.self_attn.k_bmm_quantizer
model.layers.15.self_attn.v_bmm_quantizer
model.layers.15.self_attn.softmax_quantizer
model.layers.15.mlp.gate_proj.input_quantizer
model.layers.15.mlp.gate_proj.output_quantizer
model.layers.15.mlp.gate_proj.weight_quantizer
model.layers.15.mlp.up_proj.input_quantizer
model.layers.15.mlp.up_proj.output_quantizer
model.layers.15.mlp.up_proj.weight_quantizer
model.layers.15.mlp.down_proj.input_quantizer
model.layers.15.mlp.down_proj.output_quantizer
model.layers.15.mlp.down_proj.weight_quantizer
model.layers.16.self_attn.q_proj.input_quantizer
model.layers.16.self_attn.q_proj.output_quantizer
model.layers.16.self_attn.q_proj.weight_quantizer
model.layers.16.self_attn.k_proj.input_quantizer
model.layers.16.self_attn.k_proj.output_quantizer
model.layers.16.self_attn.k_proj.weight_quantizer

model.layers.16.self_attn.v_proj.input_quantizer
model.layers.16.self_attn.v_proj.output_quantizer
model.layers.16.self_attn.v_proj.weight_quantizer
model.layers.16.self_attn.o_proj.input_quantizer
model.layers.16.self_attn.o_proj.output_quantizer
model.layers.16.self_attn.o_proj.weight_quantizer
model.layers.16.self_attn.q_bmm_quantizer
model.layers.16.self_attn.k_bmm_quantizer
model.layers.16.self_attn.v_bmm_quantizer
model.layers.16.self_attn.softmax_quantizer
model.layers.16.mlp.gate_proj.input_quantizer
model.layers.16.mlp.gate_proj.output_quantizer
model.layers.16.mlp.gate_proj.weight_quantizer
model.layers.16.mlp.up_proj.input_quantizer
model.layers.16.mlp.up_proj.output_quantizer
model.layers.16.mlp.up_proj.weight_quantizer
model.layers.16.mlp.down_proj.input_quantizer
model.layers.16.mlp.down_proj.output_quantizer
model.layers.16.mlp.down_proj.weight_quantizer
model.layers.17.self_attn.q_proj.input_quantizer
model.layers.17.self_attn.q_proj.output_quantizer
model.layers.17.self_attn.q_proj.weight_quantizer
model.layers.17.self_attn.k_proj.input_quantizer
model.layers.17.self_attn.k_proj.output_quantizer
model.layers.17.self_attn.k_proj.weight_quantizer
model.layers.17.self_attn.v_proj.input_quantizer
model.layers.17.self_attn.v_proj.output_quantizer
model.layers.17.self_attn.v_proj.weight_quantizer
model.layers.17.self_attn.o_proj.input_quantizer
model.layers.17.self_attn.o_proj.output_quantizer
model.layers.17.self_attn.o_proj.weight_quantizer
model.layers.17.self_attn.q_bmm_quantizer
model.layers.17.self_attn.k_bmm_quantizer
model.layers.17.self_attn.v_bmm_quantizer
model.layers.17.self_attn.softmax_quantizer
model.layers.17.mlp.gate_proj.input_quantizer
model.layers.17.mlp.gate_proj.output_quantizer
model.layers.17.mlp.gate_proj.weight_quantizer
model.layers.17.mlp.up_proj.input_quantizer
model.layers.17.mlp.up_proj.output_quantizer
model.layers.17.mlp.up_proj.weight_quantizer
model.layers.17.mlp.down_proj.input_quantizer
model.layers.17.mlp.down_proj.output_quantizer
model.layers.17.mlp.down_proj.weight_quantizer
model.layers.18.self_attn.q_proj.input_quantizer
model.layers.18.self_attn.q_proj.output_quantizer
model.layers.18.self_attn.q_proj.weight_quantizer
model.layers.18.self_attn.k_proj.input_quantizer

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model.layers.18.self_attn.k_proj.weight_quantizer
model.layers.18.self_attn.v_proj.input_quantizer
model.layers.18.self_attn.v_proj.output_quantizer
model.layers.18.self_attn.v_proj.weight_quantizer
model.layers.18.self_attn.o_proj.input_quantizer
model.layers.18.self_attn.o_proj.output_quantizer
model.layers.18.self_attn.o_proj.weight_quantizer
model.layers.18.self_attn.q_bmm_quantizer
model.layers.18.self_attn.k_bmm_quantizer
model.layers.18.self_attn.v_bmm_quantizer
model.layers.18.self_attn.softmax_quantizer
model.layers.18.mlp.gate_proj.input_quantizer
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model.layers.18.mlp.gate_proj.weight_quantizer
model.layers.18.mlp.up_proj.input_quantizer
model.layers.18.mlp.up_proj.output_quantizer
model.layers.18.mlp.up_proj.weight_quantizer
model.layers.18.mlp.down_proj.input_quantizer
model.layers.18.mlp.down_proj.output_quantizer
model.layers.18.mlp.down_proj.weight_quantizer
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model.layers.19.self_attn.k_proj.weight_quantizer
model.layers.19.self_attn.v_proj.input_quantizer
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model.layers.19.self_attn.o_proj.output_quantizer
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model.layers.19.self_attn.q_bmm_quantizer
model.layers.19.self_attn.k_bmm_quantizer
model.layers.19.self_attn.v_bmm_quantizer
model.layers.19.self_attn.softmax_quantizer
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model.layers.19.mlp.gate_proj.output_quantizer
model.layers.19.mlp.gate_proj.weight_quantizer
model.layers.19.mlp.up_proj.input_quantizer
model.layers.19.mlp.up_proj.output_quantizer
model.layers.19.mlp.up_proj.weight_quantizer
model.layers.19.mlp.down_proj.input_quantizer
model.layers.19.mlp.down_proj.output_quantizer
model.layers.19.mlp.down_proj.weight_quantizer
model.layers.20.self_attn.q_proj.input_quantizer
model.layers.20.self_attn.q_proj.output_quantizer

model.layers.20.self_attn.q_proj.weight_quantizer
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model.layers.20.self_attn.k_proj.output_quantizer
model.layers.20.self_attn.k_proj.weight_quantizer
model.layers.20.self_attn.v_proj.input_quantizer
model.layers.20.self_attn.v_proj.output_quantizer
model.layers.20.self_attn.v_proj.weight_quantizer
model.layers.20.self_attn.o_proj.input_quantizer
model.layers.20.self_attn.o_proj.output_quantizer
model.layers.20.self_attn.o_proj.weight_quantizer
model.layers.20.self_attn.q_bmm_quantizer
model.layers.20.self_attn.k_bmm_quantizer
model.layers.20.self_attn.v_bmm_quantizer
model.layers.20.self_attn.softmax_quantizer
model.layers.20.mlp.gate_proj.input_quantizer
model.layers.20.mlp.gate_proj.output_quantizer
model.layers.20.mlp.gate_proj.weight_quantizer
model.layers.20.mlp.up_proj.input_quantizer
model.layers.20.mlp.up_proj.output_quantizer
model.layers.20.mlp.up_proj.weight_quantizer
model.layers.20.mlp.down_proj.input_quantizer
model.layers.20.mlp.down_proj.output_quantizer
model.layers.20.mlp.down_proj.weight_quantizer
model.layers.21.self_attn.q_proj.input_quantizer
model.layers.21.self_attn.q_proj.output_quantizer
model.layers.21.self_attn.q_proj.weight_quantizer
model.layers.21.self_attn.k_proj.input_quantizer
model.layers.21.self_attn.k_proj.output_quantizer
model.layers.21.self_attn.k_proj.weight_quantizer
model.layers.21.self_attn.v_proj.input_quantizer
model.layers.21.self_attn.v_proj.output_quantizer
model.layers.21.self_attn.v_proj.weight_quantizer
model.layers.21.self_attn.o_proj.input_quantizer
model.layers.21.self_attn.o_proj.output_quantizer
model.layers.21.self_attn.o_proj.weight_quantizer
model.layers.21.self_attn.q_bmm_quantizer
model.layers.21.self_attn.k_bmm_quantizer
model.layers.21.self_attn.v_bmm_quantizer
model.layers.21.self_attn.softmax_quantizer
model.layers.21.mlp.gate_proj.input_quantizer
model.layers.21.mlp.gate_proj.output_quantizer
model.layers.21.mlp.gate_proj.weight_quantizer
model.layers.21.mlp.up_proj.input_quantizer
model.layers.21.mlp.up_proj.output_quantizer
model.layers.21.mlp.up_proj.weight_quantizer
model.layers.21.mlp.down_proj.input_quantizer
model.layers.21.mlp.down_proj.output_quantizer
model.layers.21.mlp.down_proj.weight_quantizer

model.layers.22.self_attn.q_proj.input_quantizer
model.layers.22.self_attn.q_proj.output_quantizer
model.layers.22.self_attn.q_proj.weight_quantizer
model.layers.22.self_attn.k_proj.input_quantizer
model.layers.22.self_attn.k_proj.output_quantizer
model.layers.22.self_attn.k_proj.weight_quantizer
model.layers.22.self_attn.v_proj.input_quantizer
model.layers.22.self_attn.v_proj.output_quantizer
model.layers.22.self_attn.v_proj.weight_quantizer
model.layers.22.self_attn.o_proj.input_quantizer
model.layers.22.self_attn.o_proj.output_quantizer
model.layers.22.self_attn.o_proj.weight_quantizer
model.layers.22.self_attn.q_bmm_quantizer
model.layers.22.self_attn.k_bmm_quantizer
model.layers.22.self_attn.v_bmm_quantizer
model.layers.22.self_attn.softmax_quantizer
model.layers.22.mlp.gate_proj.input_quantizer
model.layers.22.mlp.gate_proj.output_quantizer
model.layers.22.mlp.gate_proj.weight_quantizer
model.layers.22.mlp.up_proj.input_quantizer
model.layers.22.mlp.up_proj.output_quantizer
model.layers.22.mlp.up_proj.weight_quantizer
model.layers.22.mlp.down_proj.input_quantizer
model.layers.22.mlp.down_proj.output_quantizer
model.layers.22.mlp.down_proj.weight_quantizer
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model.layers.23.self_attn.q_proj.weight_quantizer
model.layers.23.self_attn.k_proj.input_quantizer
model.layers.23.self_attn.k_proj.output_quantizer
model.layers.23.self_attn.k_proj.weight_quantizer
model.layers.23.self_attn.v_proj.input_quantizer
model.layers.23.self_attn.v_proj.output_quantizer
model.layers.23.self_attn.v_proj.weight_quantizer
model.layers.23.self_attn.o_proj.input_quantizer
model.layers.23.self_attn.o_proj.output_quantizer
model.layers.23.self_attn.o_proj.weight_quantizer
model.layers.23.self_attn.q_bmm_quantizer
model.layers.23.self_attn.k_bmm_quantizer
model.layers.23.self_attn.v_bmm_quantizer
model.layers.23.self_attn.softmax_quantizer
model.layers.23.mlp.gate_proj.input_quantizer
model.layers.23.mlp.gate_proj.output_quantizer
model.layers.23.mlp.gate_proj.weight_quantizer
model.layers.23.mlp.up_proj.input_quantizer
model.layers.23.mlp.up_proj.output_quantizer
model.layers.23.mlp.up_proj.weight_quantizer
model.layers.23.mlp.down_proj.input_quantizer

model.layers.23.mlp.down_proj.output_quantizer
model.layers.23.mlp.down_proj.weight_quantizer
model.layers.24.self_attn.q_proj.input_quantizer
model.layers.24.self_attn.q_proj.output_quantizer
model.layers.24.self_attn.q_proj.weight_quantizer
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model.layers.24.self_attn.v_proj.weight_quantizer
model.layers.24.self_attn.o_proj.input_quantizer
model.layers.24.self_attn.o_proj.output_quantizer
model.layers.24.self_attn.o_proj.weight_quantizer
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model.layers.24.self_attn.k_bmm_quantizer
model.layers.24.self_attn.v_bmm_quantizer
model.layers.24.self_attn.softmax_quantizer
model.layers.24.mlp.gate_proj.input_quantizer
model.layers.24.mlp.gate_proj.output_quantizer
model.layers.24.mlp.gate_proj.weight_quantizer
model.layers.24.mlp.up_proj.input_quantizer
model.layers.24.mlp.up_proj.output_quantizer
model.layers.24.mlp.up_proj.weight_quantizer
model.layers.24.mlp.down_proj.input_quantizer
model.layers.24.mlp.down_proj.output_quantizer
model.layers.24.mlp.down_proj.weight_quantizer
model.layers.25.self_attn.q_proj.input_quantizer
model.layers.25.self_attn.q_proj.output_quantizer
model.layers.25.self_attn.q_proj.weight_quantizer
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model.layers.25.self_attn.v_proj.weight_quantizer
model.layers.25.self_attn.o_proj.input_quantizer
model.layers.25.self_attn.o_proj.output_quantizer
model.layers.25.self_attn.o_proj.weight_quantizer
model.layers.25.self_attn.q_bmm_quantizer
model.layers.25.self_attn.k_bmm_quantizer
model.layers.25.self_attn.v_bmm_quantizer
model.layers.25.self_attn.softmax_quantizer
model.layers.25.mlp.gate_proj.input_quantizer
model.layers.25.mlp.gate_proj.output_quantizer
model.layers.25.mlp.gate_proj.weight_quantizer
model.layers.25.mlp.up_proj.input_quantizer
model.layers.25.mlp.up_proj.output_quantizer

model.layers.25.mlp.up_proj.weight_quantizer
model.layers.25.mlp.down_proj.input_quantizer
model.layers.25.mlp.down_proj.output_quantizer
model.layers.25.mlp.down_proj.weight_quantizer
model.layers.26.self_attn.q_proj.input_quantizer
model.layers.26.self_attn.q_proj.output_quantizer
model.layers.26.self_attn.q_proj.weight_quantizer
model.layers.26.self_attn.k_proj.input_quantizer
model.layers.26.self_attn.k_proj.output_quantizer
model.layers.26.self_attn.k_proj.weight_quantizer
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model.layers.26.self_attn.v_proj.weight_quantizer
model.layers.26.self_attn.o_proj.input_quantizer
model.layers.26.self_attn.o_proj.output_quantizer
model.layers.26.self_attn.o_proj.weight_quantizer
model.layers.26.self_attn.q_bmm_quantizer
model.layers.26.self_attn.k_bmm_quantizer
model.layers.26.self_attn.v_bmm_quantizer
model.layers.26.self_attn.softmax_quantizer
model.layers.26.mlp.gate_proj.input_quantizer
model.layers.26.mlp.gate_proj.output_quantizer
model.layers.26.mlp.gate_proj.weight_quantizer
model.layers.26.mlp.up_proj.input_quantizer
model.layers.26.mlp.up_proj.output_quantizer
model.layers.26.mlp.up_proj.weight_quantizer
model.layers.26.mlp.down_proj.input_quantizer
model.layers.26.mlp.down_proj.output_quantizer
model.layers.26.mlp.down_proj.weight_quantizer
model.layers.27.self_attn.q_proj.input_quantizer
model.layers.27.self_attn.q_proj.output_quantizer
model.layers.27.self_attn.q_proj.weight_quantizer
model.layers.27.self_attn.k_proj.input_quantizer
model.layers.27.self_attn.k_proj.output_quantizer
model.layers.27.self_attn.k_proj.weight_quantizer
model.layers.27.self_attn.v_proj.input_quantizer
model.layers.27.self_attn.v_proj.output_quantizer
model.layers.27.self_attn.v_proj.weight_quantizer
model.layers.27.self_attn.o_proj.input_quantizer
model.layers.27.self_attn.o_proj.output_quantizer
model.layers.27.self_attn.o_proj.weight_quantizer
model.layers.27.self_attn.q_bmm_quantizer
model.layers.27.self_attn.k_bmm_quantizer
model.layers.27.self_attn.v_bmm_quantizer
model.layers.27.self_attn.softmax_quantizer
model.layers.27.mlp.gate_proj.input_quantizer
model.layers.27.mlp.gate_proj.output_quantizer
model.layers.27.mlp.gate_proj.weight_quantizer

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model.layers.27.mlp.up_proj.input_quantizer
model.layers.27.mlp.up_proj.output_quantizer
model.layers.27.mlp.up_proj.weight_quantizer
model.layers.27.mlp.down_proj.input_quantizer
model.layers.27.mlp.down_proj.output_quantizer
model.layers.27.mlp.down_proj.weight_quantizer
model.layers.28.self_attn.q_proj.input_quantizer
model.layers.28.self_attn.q_proj.output_quantizer
model.layers.28.self_attn.q_proj.weight_quantizer
model.layers.28.self_attn.k_proj.input_quantizer
model.layers.28.self_attn.k_proj.output_quantizer
model.layers.28.self_attn.k_proj.weight_quantizer
model.layers.28.self_attn.v_proj.input_quantizer
model.layers.28.self_attn.v_proj.output_quantizer
model.layers.28.self_attn.v_proj.weight_quantizer
model.layers.28.self_attn.o_proj.input_quantizer
model.layers.28.self_attn.o_proj.output_quantizer
model.layers.28.self_attn.o_proj.weight_quantizer
model.layers.28.self_attn.q_bmm_quantizer
model.layers.28.self_attn.k_bmm_quantizer
model.layers.28.self_attn.v_bmm_quantizer
model.layers.28.self_attn.softmax_quantizer
model.layers.28.mlp.gate_proj.input_quantizer
model.layers.28.mlp.gate_proj.output_quantizer
model.layers.28.mlp.gate_proj.weight_quantizer
model.layers.28.mlp.up_proj.input_quantizer
model.layers.28.mlp.up_proj.output_quantizer
model.layers.28.mlp.up_proj.weight_quantizer
model.layers.28.mlp.down_proj.input_quantizer
model.layers.28.mlp.down_proj.output_quantizer
model.layers.28.mlp.down_proj.weight_quantizer
model.layers.29.self_attn.q_proj.input_quantizer
model.layers.29.self_attn.q_proj.output_quantizer
model.layers.29.self_attn.q_proj.weight_quantizer
model.layers.29.self_attn.k_proj.input_quantizer
model.layers.29.self_attn.k_proj.output_quantizer
model.layers.29.self_attn.k_proj.weight_quantizer
model.layers.29.self_attn.v_proj.input_quantizer
model.layers.29.self_attn.v_proj.output_quantizer
model.layers.29.self_attn.v_proj.weight_quantizer
model.layers.29.self_attn.o_proj.input_quantizer
model.layers.29.self_attn.o_proj.output_quantizer
model.layers.29.self_attn.o_proj.weight_quantizer
model.layers.29.self_attn.q_bmm_quantizer
model.layers.29.self_attn.k_bmm_quantizer
model.layers.29.self_attn.v_bmm_quantizer
model.layers.29.self_attn.softmax_quantizer
model.layers.29.mlp.gate_proj.input_quantizer
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model.layers.29.mlp.gate_proj.output_quantizer
model.layers.29.mlp.gate_proj.weight_quantizer
model.layers.29.mlp.up_proj.input_quantizer
model.layers.29.mlp.up_proj.output_quantizer
model.layers.29.mlp.up_proj.weight_quantizer
model.layers.29.mlp.down_proj.input_quantizer
model.layers.29.mlp.down_proj.output_quantizer
model.layers.29.mlp.down_proj.weight_quantizer
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model.layers.30.self_attn.q_proj.weight_quantizer
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model.layers.30.self_attn.k_proj.output_quantizer
model.layers.30.self_attn.k_proj.weight_quantizer
model.layers.30.self_attn.v_proj.input_quantizer
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model.layers.30.self_attn.v_proj.weight_quantizer
model.layers.30.self_attn.o_proj.input_quantizer
model.layers.30.self_attn.o_proj.output_quantizer
model.layers.30.self_attn.o_proj.weight_quantizer
model.layers.30.self_attn.q_bmm_quantizer
model.layers.30.self_attn.k_bmm_quantizer
model.layers.30.self_attn.v_bmm_quantizer
model.layers.30.self_attn.softmax_quantizer
model.layers.30.mlp.gate_proj.input_quantizer
model.layers.30.mlp.gate_proj.output_quantizer
model.layers.30.mlp.gate_proj.weight_quantizer
model.layers.30.mlp.up_proj.input_quantizer
model.layers.30.mlp.up_proj.output_quantizer
model.layers.30.mlp.up_proj.weight_quantizer
model.layers.30.mlp.down_proj.input_quantizer
model.layers.30.mlp.down_proj.output_quantizer
model.layers.30.mlp.down_proj.weight_quantizer
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model.layers.31.self_attn.q_proj.output_quantizer
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model.layers.31.self_attn.k_proj.output_quantizer
model.layers.31.self_attn.k_proj.weight_quantizer
model.layers.31.self_attn.v_proj.input_quantizer
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model.layers.31.self_attn.v_proj.weight_quantizer
model.layers.31.self_attn.o_proj.input_quantizer
model.layers.31.self_attn.o_proj.output_quantizer
model.layers.31.self_attn.o_proj.weight_quantizer
model.layers.31.self_attn.q_bmm_quantizer
model.layers.31.self_attn.k_bmm_quantizer
model.layers.31.self_attn.v_bmm_quantizer

model.layers.31.self_attn.softmax_quantizer
model.layers.31.mlp.gate_proj.input_quantizer
model.layers.31.mlp.gate_proj.output_quantizer
model.layers.31.mlp.gate_proj.weight_quantizer
model.layers.31.mlp.up_proj.input_quantizer
model.layers.31.mlp.up_proj.output_quantizer
model.layers.31.mlp.up_proj.weight_quantizer
model.layers.31.mlp.down_proj.input_quantizer
model.layers.31.mlp.down_proj.output_quantizer
model.layers.31.mlp.down_proj.weight_quantizer
model.layers.32.self_attn.q_proj.input_quantizer
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model.layers.32.self_attn.k_proj.weight_quantizer
model.layers.32.self_attn.v_proj.input_quantizer
model.layers.32.self_attn.v_proj.output_quantizer
model.layers.32.self_attn.v_proj.weight_quantizer
model.layers.32.self_attn.o_proj.input_quantizer
model.layers.32.self_attn.o_proj.output_quantizer
model.layers.32.self_attn.o_proj.weight_quantizer
model.layers.32.self_attn.q_bmm_quantizer
model.layers.32.self_attn.k_bmm_quantizer
model.layers.32.self_attn.v_bmm_quantizer
model.layers.32.self_attn.softmax_quantizer
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model.layers.32.mlp.gate_proj.output_quantizer
model.layers.32.mlp.gate_proj.weight_quantizer
model.layers.32.mlp.up_proj.input_quantizer
model.layers.32.mlp.up_proj.output_quantizer
model.layers.32.mlp.up_proj.weight_quantizer
model.layers.32.mlp.down_proj.input_quantizer
model.layers.32.mlp.down_proj.output_quantizer
model.layers.32.mlp.down_proj.weight_quantizer
model.layers.33.self_attn.q_proj.input_quantizer
model.layers.33.self_attn.q_proj.output_quantizer
model.layers.33.self_attn.q_proj.weight_quantizer
model.layers.33.self_attn.k_proj.input_quantizer
model.layers.33.self_attn.k_proj.output_quantizer
model.layers.33.self_attn.k_proj.weight_quantizer
model.layers.33.self_attn.v_proj.input_quantizer
model.layers.33.self_attn.v_proj.output_quantizer
model.layers.33.self_attn.v_proj.weight_quantizer
model.layers.33.self_attn.o_proj.input_quantizer
model.layers.33.self_attn.o_proj.output_quantizer
model.layers.33.self_attn.o_proj.weight_quantizer
model.layers.33.self_attn.q_bmm_quantizer

model.layers.33.self_attn.k_bmm_quantizer
model.layers.33.self_attn.v_bmm_quantizer
model.layers.33.self_attn.softmax_quantizer
model.layers.33.mlp.gate_proj.input_quantizer
model.layers.33.mlp.gate_proj.output_quantizer
model.layers.33.mlp.gate_proj.weight_quantizer
model.layers.33.mlp.up_proj.input_quantizer
model.layers.33.mlp.up_proj.output_quantizer
model.layers.33.mlp.up_proj.weight_quantizer
model.layers.33.mlp.down_proj.input_quantizer
model.layers.33.mlp.down_proj.output_quantizer
model.layers.33.mlp.down_proj.weight_quantizer
model.layers.34.self_attn.q_proj.input_quantizer
model.layers.34.self_attn.q_proj.output_quantizer
model.layers.34.self_attn.q_proj.weight_quantizer
model.layers.34.self_attn.k_proj.input_quantizer
model.layers.34.self_attn.k_proj.output_quantizer
model.layers.34.self_attn.k_proj.weight_quantizer
model.layers.34.self_attn.v_proj.input_quantizer
model.layers.34.self_attn.v_proj.output_quantizer
model.layers.34.self_attn.v_proj.weight_quantizer
model.layers.34.self_attn.o_proj.input_quantizer
model.layers.34.self_attn.o_proj.output_quantizer
model.layers.34.self_attn.o_proj.weight_quantizer
model.layers.34.self_attn.q_bmm_quantizer
model.layers.34.self_attn.k_bmm_quantizer
model.layers.34.self_attn.v_bmm_quantizer
model.layers.34.self_attn.softmax_quantizer
model.layers.34.mlp.gate_proj.input_quantizer
model.layers.34.mlp.gate_proj.output_quantizer
model.layers.34.mlp.gate_proj.weight_quantizer
model.layers.34.mlp.up_proj.input_quantizer
model.layers.34.mlp.up_proj.output_quantizer
model.layers.34.mlp.up_proj.weight_quantizer
model.layers.34.mlp.down_proj.input_quantizer
model.layers.34.mlp.down_proj.output_quantizer
model.layers.34.mlp.down_proj.weight_quantizer
model.layers.35.self_attn.q_proj.input_quantizer
model.layers.35.self_attn.q_proj.output_quantizer
model.layers.35.self_attn.q_proj.weight_quantizer
model.layers.35.self_attn.k_proj.input_quantizer
model.layers.35.self_attn.k_proj.output_quantizer
model.layers.35.self_attn.k_proj.weight_quantizer
model.layers.35.self_attn.v_proj.input_quantizer
model.layers.35.self_attn.v_proj.output_quantizer
model.layers.35.self_attn.v_proj.weight_quantizer
model.layers.35.self_attn.o_proj.input_quantizer
model.layers.35.self_attn.o_proj.output_quantizer

model.layers.35.self_attn.o_proj.weight_quantizer
model.layers.35.self_attn.q_bmm_quantizer
model.layers.35.self_attn.k_bmm_quantizer
model.layers.35.self_attn.v_bmm_quantizer
model.layers.35.self_attn.softmax_quantizer
model.layers.35.mlp.gate_proj.input_quantizer
model.layers.35.mlp.gate_proj.output_quantizer
model.layers.35.mlp.gate_proj.weight_quantizer
model.layers.35.mlp.up_proj.input_quantizer
model.layers.35.mlp.up_proj.output_quantizer
model.layers.35.mlp.up_proj.weight_quantizer
model.layers.35.mlp.down_proj.input_quantizer
model.layers.35.mlp.down_proj.output_quantizer
model.layers.35.mlp.down_proj.weight_quantizer
model.layers.36.self_attn.q_proj.input_quantizer
model.layers.36.self_attn.q_proj.output_quantizer
model.layers.36.self_attn.q_proj.weight_quantizer
model.layers.36.self_attn.k_proj.input_quantizer
model.layers.36.self_attn.k_proj.output_quantizer
model.layers.36.self_attn.k_proj.weight_quantizer
model.layers.36.self_attn.v_proj.input_quantizer
model.layers.36.self_attn.v_proj.output_quantizer
model.layers.36.self_attn.v_proj.weight_quantizer
model.layers.36.self_attn.o_proj.input_quantizer
model.layers.36.self_attn.o_proj.output_quantizer
model.layers.36.self_attn.o_proj.weight_quantizer
model.layers.36.self_attn.q_bmm_quantizer
model.layers.36.self_attn.k_bmm_quantizer
model.layers.36.self_attn.v_bmm_quantizer
model.layers.36.self_attn.softmax_quantizer
model.layers.36.mlp.gate_proj.input_quantizer
model.layers.36.mlp.gate_proj.output_quantizer
model.layers.36.mlp.gate_proj.weight_quantizer
model.layers.36.mlp.up_proj.input_quantizer
model.layers.36.mlp.up_proj.output_quantizer
model.layers.36.mlp.up_proj.weight_quantizer
model.layers.36.mlp.down_proj.input_quantizer
model.layers.36.mlp.down_proj.output_quantizer
model.layers.36.mlp.down_proj.weight_quantizer
model.layers.37.self_attn.q_proj.input_quantizer
model.layers.37.self_attn.q_proj.output_quantizer
model.layers.37.self_attn.q_proj.weight_quantizer
model.layers.37.self_attn.k_proj.input_quantizer
model.layers.37.self_attn.k_proj.output_quantizer
model.layers.37.self_attn.k_proj.weight_quantizer
model.layers.37.self_attn.v_proj.input_quantizer
model.layers.37.self_attn.v_proj.output_quantizer
model.layers.37.self_attn.v_proj.weight_quantizer

model.layers.37.self_attn.o_proj.input_quantizer
model.layers.37.self_attn.o_proj.output_quantizer
model.layers.37.self_attn.o_proj.weight_quantizer
model.layers.37.self_attn.q_bmm_quantizer
model.layers.37.self_attn.k_bmm_quantizer
model.layers.37.self_attn.v_bmm_quantizer
model.layers.37.self_attn.softmax_quantizer
model.layers.37.mlp.gate_proj.input_quantizer
model.layers.37.mlp.gate_proj.output_quantizer
model.layers.37.mlp.gate_proj.weight_quantizer
model.layers.37.mlp.up_proj.input_quantizer
model.layers.37.mlp.up_proj.output_quantizer
model.layers.37.mlp.up_proj.weight_quantizer
model.layers.37.mlp.down_proj.input_quantizer
model.layers.37.mlp.down_proj.output_quantizer
model.layers.37.mlp.down_proj.weight_quantizer
model.layers.38.self_attn.q_proj.input_quantizer
model.layers.38.self_attn.q_proj.output_quantizer
model.layers.38.self_attn.q_proj.weight_quantizer
model.layers.38.self_attn.k_proj.input_quantizer
model.layers.38.self_attn.k_proj.output_quantizer
model.layers.38.self_attn.k_proj.weight_quantizer
model.layers.38.self_attn.v_proj.input_quantizer
model.layers.38.self_attn.v_proj.output_quantizer
model.layers.38.self_attn.v_proj.weight_quantizer
model.layers.38.self_attn.o_proj.input_quantizer
model.layers.38.self_attn.o_proj.output_quantizer
model.layers.38.self_attn.o_proj.weight_quantizer
model.layers.38.self_attn.q_bmm_quantizer
model.layers.38.self_attn.k_bmm_quantizer
model.layers.38.self_attn.v_bmm_quantizer
model.layers.38.self_attn.softmax_quantizer
model.layers.38.mlp.gate_proj.input_quantizer
model.layers.38.mlp.gate_proj.output_quantizer
model.layers.38.mlp.gate_proj.weight_quantizer
model.layers.38.mlp.up_proj.input_quantizer
model.layers.38.mlp.up_proj.output_quantizer
model.layers.38.mlp.up_proj.weight_quantizer
model.layers.38.mlp.down_proj.input_quantizer
model.layers.38.mlp.down_proj.output_quantizer
model.layers.38.mlp.down_proj.weight_quantizer
model.layers.39.self_attn.q_proj.input_quantizer
model.layers.39.self_attn.q_proj.output_quantizer
model.layers.39.self_attn.q_proj.weight_quantizer
model.layers.39.self_attn.k_proj.input_quantizer
model.layers.39.self_attn.k_proj.output_quantizer
model.layers.39.self_attn.k_proj.weight_quantizer
model.layers.39.self_attn.v_proj.input_quantizer

model.layers.39.self_attn.v_proj.output_quantizer
model.layers.39.self_attn.v_proj.weight_quantizer
model.layers.39.self_attn.o_proj.input_quantizer
model.layers.39.self_attn.o_proj.output_quantizer
model.layers.39.self_attn.o_proj.weight_quantizer
model.layers.39.self_attn.q_bmm_quantizer
model.layers.39.self_attn.k_bmm_quantizer
model.layers.39.self_attn.v_bmm_quantizer
model.layers.39.self_attn.softmax_quantizer
model.layers.39.mlp.gate_proj.input_quantizer
model.layers.39.mlp.gate_proj.output_quantizer
model.layers.39.mlp.gate_proj.weight_quantizer
model.layers.39.mlp.up_proj.input_quantizer
model.layers.39.mlp.up_proj.output_quantizer
model.layers.39.mlp.up_proj.weight_quantizer
model.layers.39.mlp.down_proj.input_quantizer
model.layers.39.mlp.down_proj.output_quantizer
model.layers.39.mlp.down_proj.weight_quantizer
model.layers.40.self_attn.q_proj.input_quantizer
model.layers.40.self_attn.q_proj.output_quantizer
model.layers.40.self_attn.q_proj.weight_quantizer
model.layers.40.self_attn.k_proj.input_quantizer
model.layers.40.self_attn.k_proj.output_quantizer
model.layers.40.self_attn.k_proj.weight_quantizer
model.layers.40.self_attn.v_proj.input_quantizer
model.layers.40.self_attn.v_proj.output_quantizer
model.layers.40.self_attn.v_proj.weight_quantizer
model.layers.40.self_attn.o_proj.input_quantizer
model.layers.40.self_attn.o_proj.output_quantizer
model.layers.40.self_attn.o_proj.weight_quantizer
model.layers.40.self_attn.q_bmm_quantizer
model.layers.40.self_attn.k_bmm_quantizer
model.layers.40.self_attn.v_bmm_quantizer
model.layers.40.self_attn.softmax_quantizer
model.layers.40.mlp.gate_proj.input_quantizer
model.layers.40.mlp.gate_proj.output_quantizer
model.layers.40.mlp.gate_proj.weight_quantizer
model.layers.40.mlp.up_proj.input_quantizer
model.layers.40.mlp.up_proj.output_quantizer
model.layers.40.mlp.up_proj.weight_quantizer
model.layers.40.mlp.down_proj.input_quantizer
model.layers.40.mlp.down_proj.output_quantizer
model.layers.40.mlp.down_proj.weight_quantizer
model.layers.41.self_attn.q_proj.input_quantizer
model.layers.41.self_attn.q_proj.output_quantizer
model.layers.41.self_attn.q_proj.weight_quantizer
model.layers.41.self_attn.k_proj.input_quantizer
model.layers.41.self_attn.k_proj.output_quantizer

model.layers.41.self_attn.k_proj.weight_quantizer
model.layers.41.self_attn.v_proj.input_quantizer
model.layers.41.self_attn.v_proj.output_quantizer
model.layers.41.self_attn.v_proj.weight_quantizer
model.layers.41.self_attn.o_proj.input_quantizer
model.layers.41.self_attn.o_proj.output_quantizer
model.layers.41.self_attn.o_proj.weight_quantizer
model.layers.41.self_attn.q_bmm_quantizer
model.layers.41.self_attn.k_bmm_quantizer
model.layers.41.self_attn.v_bmm_quantizer
model.layers.41.self_attn.softmax_quantizer
model.layers.41.mlp.gate_proj.input_quantizer
model.layers.41.mlp.gate_proj.output_quantizer
model.layers.41.mlp.gate_proj.weight_quantizer
model.layers.41.mlp.up_proj.input_quantizer
model.layers.41.mlp.up_proj.output_quantizer
model.layers.41.mlp.up_proj.weight_quantizer
model.layers.41.mlp.down_proj.input_quantizer
model.layers.41.mlp.down_proj.output_quantizer
model.layers.41.mlp.down_proj.weight_quantizer
model.layers.42.self_attn.q_proj.input_quantizer
model.layers.42.self_attn.q_proj.output_quantizer
model.layers.42.self_attn.q_proj.weight_quantizer
model.layers.42.self_attn.k_proj.input_quantizer
model.layers.42.self_attn.k_proj.output_quantizer
model.layers.42.self_attn.k_proj.weight_quantizer
model.layers.42.self_attn.v_proj.input_quantizer
model.layers.42.self_attn.v_proj.output_quantizer
model.layers.42.self_attn.v_proj.weight_quantizer
model.layers.42.self_attn.o_proj.input_quantizer
model.layers.42.self_attn.o_proj.output_quantizer
model.layers.42.self_attn.o_proj.weight_quantizer
model.layers.42.self_attn.q_bmm_quantizer
model.layers.42.self_attn.k_bmm_quantizer
model.layers.42.self_attn.v_bmm_quantizer
model.layers.42.self_attn.softmax_quantizer
model.layers.42.mlp.gate_proj.input_quantizer
model.layers.42.mlp.gate_proj.output_quantizer
model.layers.42.mlp.gate_proj.weight_quantizer
model.layers.42.mlp.up_proj.input_quantizer
model.layers.42.mlp.up_proj.output_quantizer
model.layers.42.mlp.up_proj.weight_quantizer
model.layers.42.mlp.down_proj.input_quantizer
model.layers.42.mlp.down_proj.output_quantizer
model.layers.42.mlp.down_proj.weight_quantizer
model.layers.43.self_attn.q_proj.input_quantizer
model.layers.43.self_attn.q_proj.output_quantizer
model.layers.43.self_attn.q_proj.weight_quantizer

model.layers.43.self_attn.k_proj.input_quantizer
model.layers.43.self_attn.k_proj.output_quantizer
model.layers.43.self_attn.k_proj.weight_quantizer
model.layers.43.self_attn.v_proj.input_quantizer
model.layers.43.self_attn.v_proj.output_quantizer
model.layers.43.self_attn.v_proj.weight_quantizer
model.layers.43.self_attn.o_proj.input_quantizer
model.layers.43.self_attn.o_proj.output_quantizer
model.layers.43.self_attn.o_proj.weight_quantizer
model.layers.43.self_attn.q_bmm_quantizer
model.layers.43.self_attn.k_bmm_quantizer
model.layers.43.self_attn.v_bmm_quantizer
model.layers.43.self_attn.softmax_quantizer
model.layers.43.mlp.gate_proj.input_quantizer
model.layers.43.mlp.gate_proj.output_quantizer
model.layers.43.mlp.gate_proj.weight_quantizer
model.layers.43.mlp.up_proj.input_quantizer
model.layers.43.mlp.up_proj.output_quantizer
model.layers.43.mlp.up_proj.weight_quantizer
model.layers.43.mlp.down_proj.input_quantizer
model.layers.43.mlp.down_proj.output_quantizer
model.layers.43.mlp.down_proj.weight_quantizer
model.layers.44.self_attn.q_proj.input_quantizer
model.layers.44.self_attn.q_proj.output_quantizer
model.layers.44.self_attn.q_proj.weight_quantizer
model.layers.44.self_attn.k_proj.input_quantizer
model.layers.44.self_attn.k_proj.output_quantizer
model.layers.44.self_attn.k_proj.weight_quantizer
model.layers.44.self_attn.v_proj.input_quantizer
model.layers.44.self_attn.v_proj.output_quantizer
model.layers.44.self_attn.v_proj.weight_quantizer
model.layers.44.self_attn.o_proj.input_quantizer
model.layers.44.self_attn.o_proj.output_quantizer
model.layers.44.self_attn.o_proj.weight_quantizer
model.layers.44.self_attn.q_bmm_quantizer
model.layers.44.self_attn.k_bmm_quantizer
model.layers.44.self_attn.v_bmm_quantizer
model.layers.44.self_attn.softmax_quantizer
model.layers.44.mlp.gate_proj.input_quantizer
model.layers.44.mlp.gate_proj.output_quantizer
model.layers.44.mlp.gate_proj.weight_quantizer
model.layers.44.mlp.up_proj.input_quantizer
model.layers.44.mlp.up_proj.output_quantizer
model.layers.44.mlp.up_proj.weight_quantizer
model.layers.44.mlp.down_proj.input_quantizer
model.layers.44.mlp.down_proj.output_quantizer
model.layers.44.mlp.down_proj.weight_quantizer
model.layers.45.self_attn.q_proj.input_quantizer

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model.layers.45.self_attn.q_proj.weight_quantizer
model.layers.45.self_attn.k_proj.input_quantizer
model.layers.45.self_attn.k_proj.output_quantizer
model.layers.45.self_attn.k_proj.weight_quantizer
model.layers.45.self_attn.v_proj.input_quantizer
model.layers.45.self_attn.v_proj.output_quantizer
model.layers.45.self_attn.v_proj.weight_quantizer
model.layers.45.self_attn.o_proj.input_quantizer
model.layers.45.self_attn.o_proj.output_quantizer
model.layers.45.self_attn.o_proj.weight_quantizer
model.layers.45.self_attn.q_bmm_quantizer
model.layers.45.self_attn.k_bmm_quantizer
model.layers.45.self_attn.v_bmm_quantizer
model.layers.45.self_attn.softmax_quantizer
model.layers.45.mlp.gate_proj.input_quantizer
model.layers.45.mlp.gate_proj.output_quantizer
model.layers.45.mlp.gate_proj.weight_quantizer
model.layers.45.mlp.up_proj.input_quantizer
model.layers.45.mlp.up_proj.output_quantizer
model.layers.45.mlp.up_proj.weight_quantizer
model.layers.45.mlp.down_proj.input_quantizer
model.layers.45.mlp.down_proj.output_quantizer
model.layers.45.mlp.down_proj.weight_quantizer
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model.layers.46.self_attn.q_proj.output_quantizer
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model.layers.46.self_attn.k_proj.input_quantizer
model.layers.46.self_attn.k_proj.output_quantizer
model.layers.46.self_attn.k_proj.weight_quantizer
model.layers.46.self_attn.v_proj.input_quantizer
model.layers.46.self_attn.v_proj.output_quantizer
model.layers.46.self_attn.v_proj.weight_quantizer
model.layers.46.self_attn.o_proj.input_quantizer
model.layers.46.self_attn.o_proj.output_quantizer
model.layers.46.self_attn.o_proj.weight_quantizer
model.layers.46.self_attn.q_bmm_quantizer
model.layers.46.self_attn.k_bmm_quantizer
model.layers.46.self_attn.v_bmm_quantizer
model.layers.46.self_attn.softmax_quantizer
model.layers.46.mlp.gate_proj.input_quantizer
model.layers.46.mlp.gate_proj.output_quantizer
model.layers.46.mlp.gate_proj.weight_quantizer
model.layers.46.mlp.up_proj.input_quantizer
model.layers.46.mlp.up_proj.output_quantizer
model.layers.46.mlp.up_proj.weight_quantizer
model.layers.46.mlp.down_proj.input_quantizer
model.layers.46.mlp.down_proj.output_quantizer

model.layers.46.mlp.down_proj.weight_quantizer
model.layers.47.self_attn.q_proj.input_quantizer
model.layers.47.self_attn.q_proj.output_quantizer
model.layers.47.self_attn.q_proj.weight_quantizer
model.layers.47.self_attn.k_proj.input_quantizer
model.layers.47.self_attn.k_proj.output_quantizer
model.layers.47.self_attn.k_proj.weight_quantizer
model.layers.47.self_attn.v_proj.input_quantizer
model.layers.47.self_attn.v_proj.output_quantizer
model.layers.47.self_attn.v_proj.weight_quantizer
model.layers.47.self_attn.o_proj.input_quantizer
model.layers.47.self_attn.o_proj.output_quantizer
model.layers.47.self_attn.o_proj.weight_quantizer
model.layers.47.self_attn.q_bmm_quantizer
model.layers.47.self_attn.k_bmm_quantizer
model.layers.47.self_attn.v_bmm_quantizer
model.layers.47.self_attn.softmax_quantizer
model.layers.47.mlp.gate_proj.input_quantizer
model.layers.47.mlp.gate_proj.output_quantizer
model.layers.47.mlp.gate_proj.weight_quantizer
model.layers.47.mlp.up_proj.input_quantizer
model.layers.47.mlp.up_proj.output_quantizer
model.layers.47.mlp.up_proj.weight_quantizer
model.layers.47.mlp.down_proj.input_quantizer
model.layers.47.mlp.down_proj.output_quantizer
model.layers.47.mlp.down_proj.weight_quantizer
model.layers.48.self_attn.q_proj.input_quantizer
model.layers.48.self_attn.q_proj.output_quantizer
model.layers.48.self_attn.q_proj.weight_quantizer
model.layers.48.self_attn.k_proj.input_quantizer
model.layers.48.self_attn.k_proj.output_quantizer
model.layers.48.self_attn.k_proj.weight_quantizer
model.layers.48.self_attn.v_proj.input_quantizer
model.layers.48.self_attn.v_proj.output_quantizer
model.layers.48.self_attn.v_proj.weight_quantizer
model.layers.48.self_attn.o_proj.input_quantizer
model.layers.48.self_attn.o_proj.output_quantizer
model.layers.48.self_attn.o_proj.weight_quantizer
model.layers.48.self_attn.q_bmm_quantizer
model.layers.48.self_attn.k_bmm_quantizer
model.layers.48.self_attn.v_bmm_quantizer
model.layers.48.self_attn.softmax_quantizer
model.layers.48.mlp.gate_proj.input_quantizer
model.layers.48.mlp.gate_proj.output_quantizer
model.layers.48.mlp.gate_proj.weight_quantizer
model.layers.48.mlp.up_proj.input_quantizer
model.layers.48.mlp.up_proj.output_quantizer
model.layers.48.mlp.up_proj.weight_quantizer

model.layers.48.mlp.down_proj.input_quantizer
model.layers.48.mlp.down_proj.output_quantizer
model.layers.48.mlp.down_proj.weight_quantizer
model.layers.49.self_attn.q_proj.input_quantizer
model.layers.49.self_attn.q_proj.output_quantizer
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lm_head.input_quantizer
lm_head.output_quantizer
lm_head.weight_quantizer
1603 TensorQuantizers found in model
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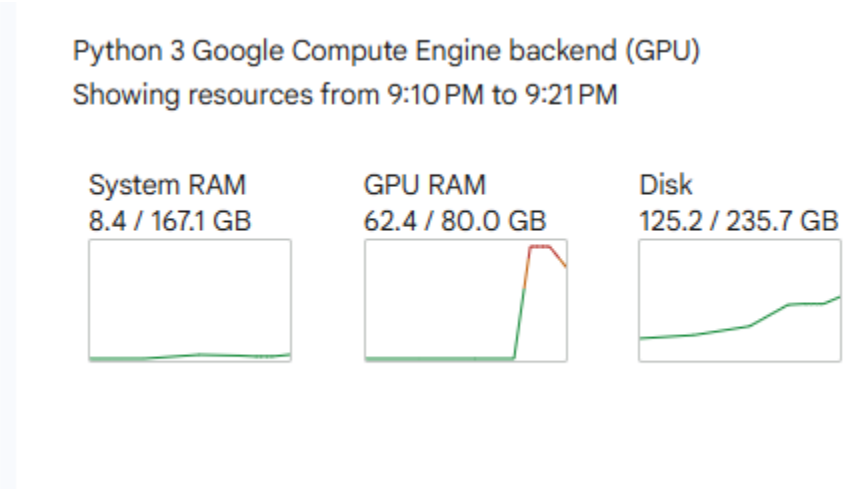
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
/usr/local/lib/python3.12/dist-packages/modelopt/torch/export/unified_export_
  weight_scaling_factor = torch.tensor(weight_quantizer.amax / weight_quantiz
```

✅ КВАНТОВАНИЕ 32B-МОДЕЛИ ЗАВЕРШЕНО УСПЕШНО!

```
--- Содержимое папки с готовой моделью ---
total 32G
-rw-r--r-- 1 root root  605 Mar 23 01:22 added_tokens.json
-rw-r--r-- 1 root root 2.5K Mar 23 01:22 chat_template.jinja
-rw-r--r-- 1 root root 3.1K Mar 23 01:22 config.json
-rw-r--r-- 1 root root  243 Mar 23 01:20 generation_config.json
-rw-r--r-- 1 root root  239 Mar 23 01:20 hf_quant_config.json
-rw-r--r-- 1 root root 1.6M Mar 23 01:22 merges.txt
-rw-r--r-- 1 root root 4.7G Mar 23 01:20 model-00001-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00002-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00003-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00004-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00005-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00006-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:22 model-00007-of-00007.safetensors
-rw-r--r-- 1 root root 138K Mar 23 01:22 model.safetensors.index.json
-rw-r--r-- 1 root root  613 Mar 23 01:22 special_tokens_map.json
-rw-r--r-- 1 root root 4.6K Mar 23 01:22 tokenizer_config.json
-rw-r--r-- 1 root root  11M Mar 23 01:22 tokenizer.json
-rw-r--r-- 1 root root 2.7M Mar 23 01:22 vocab.json
```



В момент квантизирования 32B

```
✅ КВАНТОВАНИЕ 32B-МОДЕЛИ ЗАВЕРШЕНО УСПЕШНО!

--- Содержимое папки с готовой моделью ---
total 32G
-rw-r--r-- 1 root root  605 Mar 23 01:22 added_tokens.json
-rw-r--r-- 1 root root 2.5K Mar 23 01:22 chat_template.jinja
-rw-r--r-- 1 root root 3.1K Mar 23 01:22 config.json
-rw-r--r-- 1 root root  243 Mar 23 01:20 generation_config.json
-rw-r--r-- 1 root root  239 Mar 23 01:20 hf_quant_config.json
-rw-r--r-- 1 root root 1.6M Mar 23 01:22 merges.txt
-rw-r--r-- 1 root root 4.7G Mar 23 01:20 model-00001-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00002-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00003-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00004-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00005-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:21 model-00006-of-00007.safetensors
-rw-r--r-- 1 root root 4.6G Mar 23 01:22 model-00007-of-00007.safetensors
-rw-r--r-- 1 root root 138K Mar 23 01:22 model.safetensors.index.json
-rw-r--r-- 1 root root  613 Mar 23 01:22 special_tokens_map.json
-rw-r--r-- 1 root root 4.6K Mar 23 01:22 tokenizer_config.json
-rw-r--r-- 1 root root  11M Mar 23 01:22 tokenizer.json
-rw-r--r-- 1 root root 2.7M Mar 23 01:22 vocab.json
```

image.png